ASD-STE100 SIMPLIFIED TECHNICAL ENGLISH EDITION

Linear Algebra: The Math Book

Equations, Plus-Two Derivations, SVG Diagrams, and Complete Verbatim Solutions

Syllabus Coverage (Gilbert Strang, 4th Edition):

- **Chapter 2:** Solving Linear Equations (Sections 2.1 – 2.7, pp. 31–119)
- **Chapter 3:** Complete Solution to $A\mathbf{x} = \mathbf{b}$ & Four Subspaces (Sections 3.4 & 3.6, pp. 155–167, 193–194)
- **Chapter 5:** Determinants, Permutations, Cofactors, Cramer's Rule, and Volumes (Sections 5.1 – 5.3, pp. 244–282)

Written in strict compliance with ASD-STE100 Simplified Technical English.

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Plus-Two (+2) Base Toolkit

We derive every equation in this book from these eight Plus-Two (Grade 11–12) mathematics facts. Refer to tags **T1**–**T8** in the derivations.

T1 Simultaneous Linear Equations in Two Unknowns

Given two lines $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, multiply the first equation by a_2/a_1 and subtract it from the second equation:

$$\left(b_2 - \frac{a_2}{a_1}b_1\right) y = c_2 - \frac{a_2}{a_1}c_1 \implies (a_1b_2 - a_2b_1)y = a_1c_2 - a_2c_1.$$

T2 Vector Addition and Scalar Multiplication

For column vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ and scalar $c \in \mathbb{R}$:

$$c\mathbf{u} = \begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}, \qquad \mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}.$$

T3 Dot Product (Inner Product) and Perpendicularity

The dot product of $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ is:

$$\mathbf{x} \cdot \mathbf{y} = \sum_{j=1}^{n} x_j y_j = x_1y_1 + x_2y_2 + \cdots + x_ny_n = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos\theta.$$

Two nonzero vectors are perpendicular (orthogonal) if and only if $\mathbf{x} \cdot \mathbf{y} = 0$.

T4 Sum of First n Integers and Squares

From +2 mathematical induction:

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{1}{3}n\left(n + \frac{1}{2}\right)(n+1) \approx \frac{1}{3}n^3 \quad \text{(for large } n\text{)}.$$

T5 Binomial Theorem and Combinations

The number of ways to choose j items from i items is $\binom{i}{j} = \frac{i!}{j!(i-j)!}$. Pascal's identity states:

$$\binom{i}{j} + \binom{i}{j-1} = \binom{i+1}{j}, \qquad (1+x)^i = \sum_{j=0}^{i} \binom{i}{j} x^j.$$

T6 2 by 2 Determinant and Area of a Triangle

For a 2×2 array, $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$. The area of a triangle with vertices $(0,0)$, (x_1, y_1), and (x_2, y_2) is $\frac{1}{2}|x_1y_2 - x_2y_1|$.

T7 Integration by Parts

For differentiable functions $x(t)$ and $y(t)$ that vanish at $\pm\infty$:

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\mathrm{d}t} y(t)\,\mathrm{d}t = \Big[x(t)y(t)\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x(t)\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t = \int_{-\infty}^{\infty} x(t)\left(-\frac{\mathrm{d}y}{\mathrm{d}t}\right)\,\mathrm{d}t.$$

T8 3D Vector Cross Product

In $\mathbb{R}^3$, the cross product $\mathbf{u} \times \mathbf{v}$ has magnitude $\|\mathbf{u}\|\,\|\mathbf{v}\|\,|\sin\theta|$ and points perpendicular to both $\mathbf{u}$ and $\mathbf{v}$ by the right-hand rule.

CHAPTER 2

Solving Linear Equations

Chapter 2 Syllabus: 2.1 Vectors and Linear Equations · 2.2 The Idea of Elimination · 2.3 Elimination Using Matrices · 2.4 Rules for Matrix Operations · 2.5 Inverse Matrices · 2.6 Elimination = Factorization: $A = LU$ · 2.7 Transposes and Permutations.

2.1 Vectors and Linear Equations [Book 2.1, pp. 31–44]

2.1.1 Derivation 1: Row Picture, Column Picture, and Matrix Form $A\mathbf{x} = \mathbf{b}$

GOAL

Write a system of m linear equations in n unknowns in three equivalent forms: Row Equations, Vector Linear Combination, and Matrix Equation $A\mathbf{x} = \mathbf{b}$.

FROM YOUR +2 BASE

Use **T1** (simultaneous equations), **T2** (vector operations), and **T3** (dot product).

Step 1: Start from the 2×2 linear system. Consider the two equations in two unknowns x and y [Book: Eq. (1), p. 31]:

$$x - 2y = 1 \tag{2.1}$$
$$3x + 2y = 11. \tag{2.2}$$

In the **row picture**, each equation is a line in the xy-plane. The two lines meet at $(x, y) = (3, 1)$.

Step 2: Separate the system into columns. Group the coefficients of x, the coefficients of y, and the right-hand side numbers into column vectors. Apply **T2** to get the **column picture** [Book: Eq. (2), p. 32]:

$$x\begin{bmatrix}1\\3\end{bmatrix} + y\begin{bmatrix}-2\\2\end{bmatrix} = \begin{bmatrix}1x + (-2)y\\3x + 2y\end{bmatrix} = \begin{bmatrix}1\\11\end{bmatrix} = \mathbf{b}. \tag{2.3}$$

For $(x, y) = (3, 1)$, scalar multiplication and vector addition give:

$$3\begin{bmatrix}1\\3\end{bmatrix} + 1\begin{bmatrix}-2\\2\end{bmatrix} = \begin{bmatrix}3\\9\end{bmatrix} + \begin{bmatrix}-2\\2\end{bmatrix} = \begin{bmatrix}1\\11\end{bmatrix} = \mathbf{b}. \tag{2.4}$$

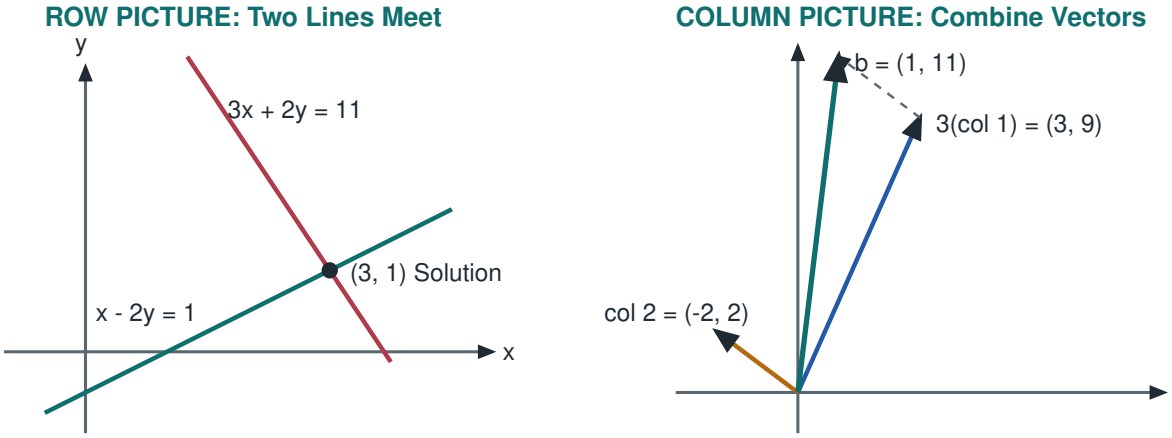


Fig. 2.1 Left: Row picture shows two lines that meet at $(3,1)$. Right: Column picture combines $3(\text{column } 1) + 1(\text{column } 2)$ to produce $\mathbf{b} = (1,11)$.

Step 3: Extend to three equations in three unknowns. Write the 3×3 system [Book: Eq. (3), p. 34]:

$$\begin{aligned} x + 2y + 3z &= 6 \\ 2x + 5y + 2z &= 4 \\ 6x - 3y + z &= 2. \end{aligned} \tag{2.5}$$

Separate the columns to get the vector equation [Book: Eq. (4), p. 35]:

$$x\begin{bmatrix}1\\2\\6\end{bmatrix} + y\begin{bmatrix}2\\5\\-3\end{bmatrix} + z\begin{bmatrix}3\\2\\1\end{bmatrix} = \begin{bmatrix}6\\4\\2\end{bmatrix} = \mathbf{b}. \tag{2.6}$$

By inspection, $0(\text{column } 1) + 0(\text{column } 2) + 2(\text{column } 3) = (6,4,2)$, so $(x,y,z) = (0,0,2)$.

Step 4: Pack the columns into the coefficient matrix A. Place the three columns into a 3×3 matrix A and the unknowns into a column vector $\mathbf{x}$ [Book: Eq. (5), p. 36]:

$$A\mathbf{x} = \begin{bmatrix}1 & 2 & 3\\2 & 5 & 2\\6 & -3 & 1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}6\\4\\2\end{bmatrix} = \mathbf{b}. \tag{2.7}$$

RESULTMatrix-Vector Multiplication Rules [Book Eqs. (6)–(7), p. 36]

We multiply A times $\mathbf{x}$ in two equivalent ways:

1. By Rows: $$A\mathbf{x} = \begin{bmatrix}(\text{row } 1)\cdot\mathbf{x}\\(\text{row } 2)\cdot\mathbf{x}\\(\text{row } 3)\cdot\mathbf{x}\end{bmatrix}, \qquad (A\mathbf{x})_i = \sum_{j=1}^{n} a_{ij}x_j, \tag{2.8}$$

2. By Columns: $$A\mathbf{x} = x_1(\text{column } 1) + x_2(\text{column } 2) + \cdots + x_n(\text{column } n). \tag{2.9}$$

The $n \times n$ **identity matrix** I has 1's on the main diagonal and 0's elsewhere. It obeys $\boxed{I\mathbf{x} = \mathbf{x}}$.

2.1.2 Verbatim Book Examples and Worked Examples (Section 2.1)

BOOK EXAMPLE 2.1-Ex1 Multiplication by A and Identity I

Here are 3 by 3 matrices A and I = identity, with three 1's and six 0's:

$$A\mathbf{x} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} \qquad I\mathbf{x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

BOOK EXAMPLE 2.1 A Column Picture Inspection

Describe the column picture of these three equations $A\mathbf{x} = \mathbf{b}$. Solve by careful inspection of the columns (instead of elimination):

$$\begin{aligned} x + 3y + 2z &= -3 \\ 2x + 2y + 2z &= -2 \\ 3x + 5y + 6z &= -5 \end{aligned} \quad \text{which is} \quad \begin{bmatrix} 1 & 3 & 2 \\ 2 & 2 & 2 \\ 3 & 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \\ -5 \end{bmatrix}.$$

Suppose the right side changes to $\mathbf{b} = (4, 4, 8)$ = sum of the first two columns. What is the new solution?

BOOK EXAMPLE 2.1 B Singular System with No Solution

This system has *no solution*. The planes in the row picture don't meet at a point. *No combination of the three columns produces* $\mathbf{b}$. How to show this?

$$\begin{aligned} x + 3y + 5z &= 4 \\ x + 2y - 3z &= 5 \\ 2x + 5y + 2z &= 8 \end{aligned} \qquad \begin{bmatrix} 1 & 3 & 5 \\ 1 & 2 & -3 \\ 2 & 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix} = \mathbf{b}.$$

(1) Multiply the equations by $1, 1, -1$ and add to get $0 = 1$. *No solution*. Are any two of the planes parallel? What are the equations of planes parallel to $x + 3y + 5z = 4$?
(2) Take the dot product of each column of A (and also $\mathbf{b}$) with $\mathbf{y} = (1, 1, -1)$. How do those dot products show that the system $A\mathbf{x} = \mathbf{b}$ has no solution?
(3) Find three right side vectors $\mathbf{b}^*$ and $\mathbf{b}^{**}$ and $\mathbf{b}^{***}$ that *do* allow solutions.

SOLUTION AT END OF PAGE (Examples 2.1-Ex1, 2.1 A, and 2.1 B)

2.1-Ex1: By rows, the dot product of $(1, 0, 0)$ with $(4, 5, 6)$ is 4. By columns, $A\mathbf{x}$ is 4 times column 1, which gives $4(1, 1, 1) = (4, 4, 4)$. For I, $I\mathbf{x} = 4(1, 0, 0) + 5(0, 1, 0) + 6(0, 0, 1) = (4, 5, 6) = \mathbf{x}$.

2.1 A: The vector $\mathbf{b} = (-3, -2, -5)$ is minus the second column $(3, 2, 5)$. Thus $x = 0, y = -1, z = 0$, or $\mathbf{x} = (0, -1, 0)$. Elimination gives three nonzero pivots, so this solution is unique. When $\mathbf{b} = (4, 4, 8) = (\text{column } 1) + (\text{column } 2)$, the weights are $x = 1, y = 1, z = 0$, so $\mathbf{x} = (1, 1, 0)$.

2.1 B: (1) $(x + 3y + 5z) + (x + 2y - 3z) - (2x + 5y + 2z) = 0x + 0y + 0z$, while $4 + 5 - 8 = 1$, which gives $0 = 1$ (no solution). No two planes are parallel because their normal vectors are not multiples. Planes parallel to $x + 3y + 5z = 4$ have the equation $x + 3y + 5z = c$ (for example $x + 3y + 5z = 0$).

(2) Dot products with $\mathbf{y} = (1, 1, -1)$: $\mathbf{y} \cdot (\text{col } 1) = 1 + 1 - 2 = 0$, $\mathbf{y} \cdot (\text{col } 2) = 3 + 2 - 5 = 0$, $\mathbf{y} \cdot (\text{col } 3) = 5 - 3 - 2 = 0$. Every combination of the columns has zero dot product with $\mathbf{y}$, but $\mathbf{y} \cdot \mathbf{b} = 4 + 5 - 8 = 1 \neq 0$. Thus no solution exists.
(3) Choose combinations of the columns: $\mathbf{b}^* = \text{col } 1 = (1, 1, 2)$ (solved by $\mathbf{x}^* = (1, 0, 0)$); $\mathbf{b}^{**} = \text{sum of cols} = (9, 0, 9)$ (solved by $\mathbf{x}^{**} = (1, 1, 1)$); $\mathbf{b}^{***} = (0, 0, 0)$ (solved by $\mathbf{x}^{***} = (0, 0, 0)$).

2.1.3 Problem Set 2.1: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 2.1 (1–8) Row and Column Pictures of $Ax = b$

1. With $A = I$ (the identity matrix) draw the planes in the row picture. Three sides of a box meet at the solution $\mathbf{x} = (x, y, z) = (2, 3, 4)$: $1x + 0y + 0z = 2$, $0x + 1y + 0z = 3$, $0x + 0y + 1z = 4$. Draw the vectors in the column picture. Two times column 1 plus three times column 2 plus four times column 3 equals the right side $\mathbf{b}$.
2. If the equations in Problem 1 are multiplied by $2, 3, 4$ they become $D\mathbf{X} = \mathbf{B}$: $2x = 4, 3y = 9, 4z = 16$. Why is the row picture the same? Is the solution $\mathbf{X}$ the same as $\mathbf{x}$? What is changed in the column picture—the columns or the right combination to give $\mathbf{B}$?
3. If equation 1 is added to equation 2, which of these are changed: the planes in the row picture, the vectors in the column picture, the coefficient matrix, the solution? The new equations in Problem 1 would be $x = 2, x + y = 5, z = 4$.
4. Find a point with $z = 2$ on the intersection line of the planes $x + y + 3z = 6$ and $x - y + z = 4$. Find the point with $z = 0$. Find a third point halfway between.
5. The first of these equations plus the second equals the third: $x + y + z = 2$, $x + 2y + z = 3$, $2x + 3y + 2z = 5$. The first two planes meet along a line. The third plane contains that line, because if x, y, z satisfy the first two equations then they also _____. The equations have infinitely many solutions (the whole line $\mathbf{L}$). Find three solutions on $\mathbf{L}$.
6. Move the third plane in Problem 5 to a parallel plane $2x + 3y + 2z = 9$. Now the three equations have no solution—why not? The first two planes meet along the line $\mathbf{L}$, but the third plane doesn't _____ that line.
7. In Problem 5 the columns are $(1, 1, 2)$ and $(1, 2, 3)$ and $(1, 1, 2)$. This is a "singular case" because the third column is _____. Find two combinations of the columns that give $\mathbf{b} = (2, 3, 5)$. This is only possible for $\mathbf{b} = (4, 6, c)$ if $c =$ _____.
8. Normally 4 "planes" in 4-dimensional space meet at a _____. Normally 4 column vectors in 4-dimensional space can combine to produce $\mathbf{b}$. What combination of $(1, 0, 0, 0), (1, 1, 0, 0), (1, 1, 1, 0), (1, 1, 1, 1)$ produces $\mathbf{b} = (3, 3, 3, 2)$? What 4 equations for x, y, z, t are you solving?

SOLUTION AT END OF PAGE (Problem Set 2.1 (Problems 1–8))

1. Row picture: planes $x = 2, y = 3, z = 4$ parallel to the coordinate planes meet at $(2, 3, 4)$. Column picture: $2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k} = (2, 3, 4)$.
2. Multiplying an equation by a nonzero number does not change its plane. The solution $\mathbf{X} = (2, 3, 4) = \mathbf{x}$ is unchanged. In the column picture, the columns change to $(2, 0, 0), (0, 3, 0), (0, 0, 4)$ and $\mathbf{B} = (4, 9, 16)$, while the combination weights $(2, 3, 4)$ stay the same.
3. Changed: the second plane in the row picture, the column vectors, and the coefficient matrix. Unchanged: the solution $(2, 3, 4)$.
4. For $z = 2$: $x + y = 0$ and $x - y = 2 \implies (x, y, z) = (1, -1, 2)$. For $z = 0$: $x + y = 6$ and $x - y = 4 \implies (5, 1, 0)$. Halfway point: $\frac{1}{2}((1, -1, 2) + (5, 1, 0)) = (3, 0, 1)$.
5. Fill in: **satisfy the third equation**. Subtract eq. 1 from eq. 2 to get $y = 1$, so $x + z = 1$. Three solutions on $\mathbf{L}$: $(1, 1, 0)$, $(0, 1, 1)$, and $(2, 1, -1)$.
6. Eq. 1 + Eq. 2 gives $2x + 3y + 2z = 5 \neq 9$. Fill in: **intersect (or contain)**.
7. Fill in: **equal to column 1**. Combinations for $\mathbf{b} = (2, 3, 5)$: $(x, y, z) = (1, 1, 0)$ and $(0, 1, 1)$. Since $b_3 = b_1 + b_2$, for $\mathbf{b} = (4, 6, c)$ we need $c = 4 + 6 =$ **10**.
8. Fill in: **point**. The equations are $x + y + z + t = 3$, $y + z + t = 3$, $z + t = 3$, $t = 2$. Back

substitution gives $t = 2, z = 1, y = 0, x = 0$.

BOOK EXAMPLE PS 2.1 (9–22) Multiplying Matrices and Special Transformations

9. Compute each $A\mathbf{x}$ by dot products of the rows with the column vector: (a) $\begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}$.

10. Compute each $A\mathbf{x}$ in Problem 9 as a combination of the columns. How many separate multiplications for $A\mathbf{x}$, when the matrix is "3 by 3"?

11. Find the two components of $A\mathbf{x}$ by rows or by columns: $\begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$.

12. Multiply A times $\mathbf{x}$ to find three components of $A\mathbf{x}$: $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

13. (a) A matrix with m rows and n columns multiplies a vector with ____ components to produce a vector with ____ components. (b) The planes from the m equations $A\mathbf{x} = \mathbf{b}$ are in ____-dimensional space. The combination of the columns of A is in ____-dimensional space.

14. Write $2x + 3y + z + 5t = 8$ as a matrix A (how many rows?) multiplying the column vector $\mathbf{x} = (x, y, z, t)$ to produce $\mathbf{b}$.

15. (a) What is the 2 by 2 identity matrix? (b) What is the 2 by 2 exchange matrix P such that $P \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$?

16. (a) What 2 by 2 matrix R rotates every vector by 90° so $R \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$? (b) What 2 by 2 matrix R^2 rotates every vector by 180°?

17. Find the matrix P that multiplies (x, y, z) to give (y, z, x). Find the matrix Q that multiplies (y, z, x) to bring back (x, y, z).

18. What 2 by 2 matrix E subtracts the first component from the second component? What 3 by 3 matrix does the same?

19. What 3 by 3 matrix E multiplies (x, y, z) to give $(x, y, z + x)$? What matrix E^{-1} multiplies (x, y, z) to give $(x, y, z - x)$? If you multiply $(3, 4, 5)$ by E and then multiply by E^{-1}, the two results are (____) and (____).

20. What 2 by 2 matrix P_1 projects (x, y) onto the x axis to produce $(x, 0)$? What matrix P_2 projects onto the y axis to produce $(0, y)$? If you multiply $(5, 7)$ by P_1 and then multiply by P_2, you get (____) and (____).

21. What 2 by 2 matrix R rotates every vector through 45°?

22. Write the dot product of $(1, 4, 5)$ and (x, y, z) as a matrix multiplication $A\mathbf{x}$. The solutions to $A\mathbf{x} = 0$ lie on a ____ perpendicular to the vector ____. The columns of A are only in ____-dimensional space.

SOLUTION AT END OF PAGE (Problem Set 2.1 (Problems 9–22))

9. (a) $(2+4+12, -4+6+3, -8+2+6) = (18,5,0)$. (b) $(2+1, 1+2+1, 1+2+2, 1+4) = (3,4,5,5)$.

10. (a) $2(1,-2,-4) + 2(2,3,1) + 3(4,1,2) = (18,5,0)$. (b) $1(2,1,0,0) + 1(1,2,1,0) + 1(0,1,2,1) + 2(0,0,1,2) = (3,4,5,5)$. A 3×3 matrix uses 9 separate multiplications.

11. $(14,22)$, $(0,0)$, and $(9,7)$. **12.** (z,y,x), $(0,0,0)$, and $(3,3,6)$.

13. (a) n, m. (b) n-dimensional space, m-dimensional space.

14. $A = \begin{bmatrix} 2 & 3 & 1 & 5 \end{bmatrix}$ (1 row), $\mathbf{b} = [8]$. **15.** (a) $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. (b) $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

16. (a) $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. (b) $R^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$.

17. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. **18.** $E_{2\times 2} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, $E_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

19. $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$. Results: $(3,4,8)$ and $(3,4,5)$.

20. $P_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $P_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Results: $(5,0)$ and $(0,0)$.

21. $R = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$. **22.** $A = \begin{bmatrix} 1 & 4 & 5 \end{bmatrix}$. Fill in: **plane**, $(1,4,5)$, 1-dimensional space.

BOOK EXAMPLE PS 2.1 (23–35) MATLAB, Markov Matrices, and Challenge Problems

23. In MATLAB notation, write the commands that define $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$. What command would test whether or not $A\mathbf{x} = \mathbf{b}$?

24. The MATLAB commands `A = eye(3)` and `v = [3:5]'` produce the 3 by 3 identity matrix and the column vector $(3,4,5)$. What are the outputs from `A*v` and `v'*v`? If you ask for `v*A`, what happens?

25. If you multiply the 4 by 4 all-ones matrix `A = ones(4)` and the column `v = ones(4,1)`, what is `A*v`? If you multiply `B = eye(4) + ones(4)` times `w = zeros(4,1) + 2*ones(4,1)`, what is `B*w`?

26. Draw the row and column pictures for the equations $x - 2y = 0, x + y = 6$.

27. For two linear equations in three unknowns x, y, z, the row picture will show (2 or 3) (lines or planes) in (2 or 3)-dimensional space. The column picture is in (2 or 3)-dimensional space. The solutions normally lie on a ____.

28. For four linear equations in two unknowns x and y, the row picture shows four ____. The column picture is in ____-dimensional space. The equations have no solution unless the vector on the right side is a combination of ____.

29. Start with the vector $\mathbf{u}_0 = (1,0)$. Multiply again and again by the same "Markov matrix" $A = [.8\ .3; .2\ .7]$. Find $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. What property do you notice for all four vectors $\mathbf{u}_0, \mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$?

30. Continue Problem 29 from $\mathbf{u}_0 = (1,0)$ to $\mathbf{u}_7$, and from $\mathbf{v}_0 = (0,1)$ to $\mathbf{v}_7$. Guess the steady state vector $\mathbf{s}$ and check that $A\mathbf{s} = \mathbf{s}$.

31. Invent a 3 by 3 magic matrix M_3 with entries $1, 2, \ldots, 9$ where all rows, columns, and diagonals add to 15. What is M_3 times $(1,1,1)$? What is M_4 times $(1,1,1,1)$ if a 4 by 4 magic matrix has entries $1, \ldots, 16$?

32. Suppose $\mathbf{u}$ and $\mathbf{v}$ are the first two columns of a 3 by 3 matrix A. Which third columns $\mathbf{w}$ would make this matrix singular?

33. If $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ then $A\mathbf{u}$ and $A\mathbf{v}$ are the columns of A. If $\mathbf{w} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$ how is $A\mathbf{w}$ connected to $A\mathbf{u}$ and $A\mathbf{v}$?

34. Start from the four equations $-x_{i+1} + 2x_i - x_{i-1} = i$ (for $i = 1,2,3,4$ with $x_0 = x_5 = 0$). Write those equations in their matrix form $A\mathbf{x} = \mathbf{b}$. Can you solve them for x_1, x_2, x_3, x_4?

35. A 9 by 9 Sudoku matrix S has the numbers $1, \ldots, 9$ in every row and column, and in every 3 by 3 block. For the all-ones vector $\mathbf{x} = (1, \ldots, 1)$, what is $S\mathbf{x}$? Which row exchanges will produce another Sudoku matrix?

SOLUTION AT END OF PAGE (Problem Set 2.1 (Problems 23–35))

23. `A=[1 2; 3 4]; x=[5; -2]; b=[1; 7]; A*x - b` (returns zero vector).

24. `A*v` gives $(3,4,5)^T$. `v'*v` gives $9 + 16 + 25 = 50$. `v*A` gives a dimension mismatch error (3×1 times 3×3).

25. `A*v` $= (4,4,4,4)^T$. Since B hasrow sum 5 and $\mathbf{w} = (2,2,2,2)^T$, `B*w` $= (10,10,10,10)^T$.

26. Row picture: lines $x - 2y = 0$ and $x + y = 6$ meet at $(4,2)$. Column picture: $4(1,1) + 2(-2,1) = (0,6)$.

27. 2 planes in **3**-dimensional space; column picture in **2**-dimensional space; solutions lie on a **line**.

28. Four **lines** in 2D; column picture in **4**-dimensional space; combination of **the two columns**.
29. $\mathbf{u}_1 = (.8, .2)$, $\mathbf{u}_2 = (.7, .3)$, $\mathbf{u}_3 = (.65, .35)$. The components of each vector add to 1.
30. Both approach the steady-state vector $\mathbf{s} = (0.6, 0.4)^T = (\frac{3}{5}, \frac{2}{5})^T$, which satisfies $A\mathbf{s} = \mathbf{s}$.
31. $M_3(1,1,1)^T = (15,15,15)^T$ (with $M_3 = [8\ 3\ 4; 1\ 5\ 9; 6\ 7\ 2]$). Since $1 + \cdots + 16 = 136$ and $136/4 = 34$, $M_4(1,1,1,1)^T = (34,34,34,34)^T$.
32. Any linear combination $\mathbf{w} = c\mathbf{u} + d\mathbf{v}$ (lying in the plane of $\mathbf{u}$ and $\mathbf{v}$) makes A singular.
33. By linearity, $\mathbf{w} = 5\mathbf{u} + 7\mathbf{v} \implies A\mathbf{w} = 5A\mathbf{u} + 7A\mathbf{v}$.
34. $A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. Solution: $\mathbf{x} = (4,7,8,6)^T$.
35. Since $1 + \cdots + 9 = 45$, $S\mathbf{x} = (45, 45, \ldots, 45)^T$. Exchanging rows within any band of 3 rows (rows 1–3, 4–6, or 7–9) or exchanging entire 3-row bands preserves the Sudoku blocks.

2.2 The Idea of Elimination [Book 2.2, pp. 45–55]

2.2.1 Derivation 2: Gaussian Elimination, Pivots, Multipliers, and Breakdown

GOAL

Derive the multiplier formula ℓ_{ij} that transforms a linear system $A\mathbf{x} = \mathbf{b}$ into an upper triangular system $U\mathbf{x} = \mathbf{c}$ for back substitution, and classify the three breakdown cases.

FROM YOUR +2 BASE

Use **T1** (elimination of a variable between two equations).

Step 1: Eliminate x from equation 2. Start with the general 2×2 system:

$$a_{11}x + a_{12}y = b_1 \qquad (\text{Pivot } d_1 = a_{11} \neq 0) \tag{2.10}$$

$$a_{21}x + a_{22}y = b_2. \tag{2.11}$$

To make the x-coefficient in row 2 equal to zero, multiply row 1 by the **multiplier** ℓ_{21} and subtract from row 2:

$$(a_{21} - \ell_{21}a_{11})x + (a_{22} - \ell_{21}a_{12})y = b_2 - \ell_{21}b_1. \tag{2.12}$$

Set $a_{21} - \ell_{21}a_{11} = 0$ to find the multiplier and the second pivot d_2:

$$\boxed{\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{\text{entry to eliminate in row 2}}{\text{pivot in row 1}}}, \qquad \boxed{d_2 = a_{22} - \ell_{21}a_{12} = a_{22} - \frac{a_{21}a_{12}}{a_{11}}}. \tag{2.13}$$

Step 2: Generalize to $n \times n$ systems. In column j, the **pivot** is the first nonzero entry in row j. To eliminate the entry in row i (where $i > j$) below the pivot, subtract ℓ_{ij} times equation j from equation i:

$$\boxed{\ell_{ij} = \frac{\text{entry to eliminate in row } i}{\text{pivot in row } j}}. \tag{2.14}$$

This reduces $A\mathbf{x} = \mathbf{b}$ to an upper triangular system $U\mathbf{x} = \mathbf{c}$, which we solve from bottom to top by **back substitution**.

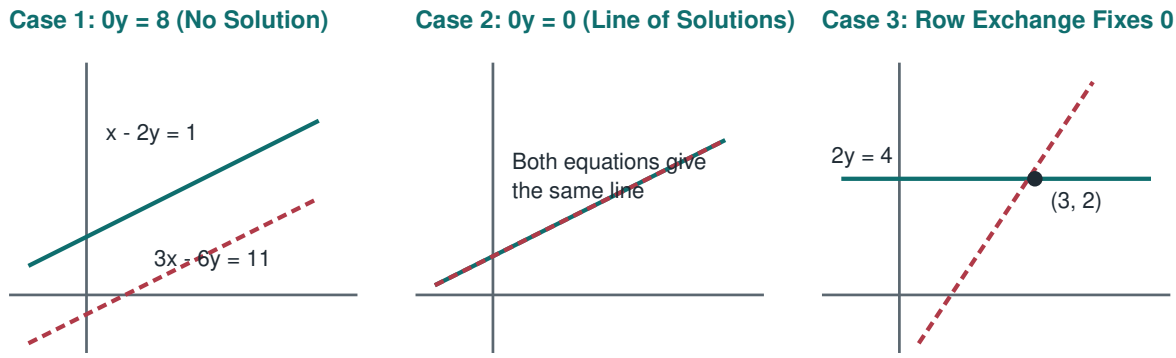


Fig. 2.2 Breakdown of elimination: Case 1 (no solution, $0 \neq 0$), Case 2 (infinitely many solutions, $0 = 0$), and Case 3 (zero in pivot position, fixed by a row exchange).

RESULTSuccess vs. Failure of Elimination [Book p. 47]

- **Nonsingular (Success):** Elimination produces n nonzero pivots (possibly after row exchanges). There is **exactly one solution**.
- **Temporary Failure:** A zero appears in a pivot position, but a nonzero entry exists below it in the same column. **Fix:** Exchange the two rows.
- **Permanent Failure (Singular):** We cannot get n pivots. Elimination reaches $0 = c \neq 0$ (**no solution**) or $0 = 0$ (**infinitely many solutions**).

2.2.2 Verbatim Book Examples and Worked Examples (Section 2.2)

BOOK EXAMPLE 2.2-Ex1–3 Three Breakdown Examples

Example 1 (Permanent failure with no solution): $x - 2y = 1$, $3x - 6y = 11$.
Example 2 (Failure with infinitely many solutions): $x - 2y = 1$, $3x - 6y = 3$.
Example 3 (Temporary failure, zero in pivot): $0x + 2y = 4$, $3x - 2y = 5$.

BOOK EXAMPLE 2.2 A Band Matrix Elimination

When elimination is applied to this matrix A, what are the first and second pivots? What is the multiplier ℓ_{21} in the first step (ℓ_{21} times row 1 is *subtracted* from row 2)?

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 2 \end{bmatrix} \longrightarrow U = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

What entry in the $2,2$ position (instead of 2) would force an exchange of rows 2 and 3? Why is the lower left multiplier $\ell_{31} = 0$, subtracting zero times row 1 from row 3? **If you change the corner entry from $a_{33} = 2$ to $a_{33} = 1$, why does elimination fail?**

BOOK EXAMPLE 2.2 B Triangular Matrix Pivots

Suppose A is already a **triangular matrix** (upper triangular or lower triangular). *Where do you see its pivots? When does* $A\mathbf{x} = \mathbf{b}$ *have exactly one solution for every* $\mathbf{b}$*?*

BOOK EXAMPLE 2.2 C Success and Failure in 3 by 3 Elimination

Use elimination to reach upper triangular matrices U. Solve by back substitution or explain why this is impossible. What are the pivots (never zero)? Exchange equations when necessary. The only difference is the $-x$ in the last equation.

$$\textbf{Success:} \quad \begin{aligned} x + y + z &= 7 \\ x + y - z &= 5 \\ x - y + z &= 3 \end{aligned} \qquad \textbf{Failure:} \quad \begin{aligned} x + y + z &= 7 \\ x + y - z &= 5 \\ -x - y + z &= 3. \end{aligned}$$

SOLUTION AT END OF PAGE (Examples 2.2-Ex1–3, 2.2 A, 2.2 B, and 2.2 C)

2.2-Ex1–3: In Ex. 1, subtracting $3(\text{row } 1)$ from row 2 gives $0y = 8$ (no second pivot, no solution). In Ex. 2, subtracting $3(\text{row } 1)$ gives $0y = 0$ (y is free, $x = 1 + 2y$, infinitely many solutions). In Ex. 3, exchange rows 1 and 2 to get $3x - 2y = 5$ and $2y = 4$; pivots are 3 and 2, giving $y = 2, x = 3$.

2.2 A: The first pivot is 1. The multiplier is $\ell_{21} = -1/1 = -1$. Subtracting $-1(\text{row } 1)$ from row 2 gives the second pivot 1. Changing $a_{22} = 2$ to 1 puts 0 in the $(2,2)$ pivot position and forces a row exchange. $\ell_{31} = 0$ because $a_{31} = 0$. If a_{33} is reduced from 2 to 1, the third pivot becomes 0 and elimination fails.

2.2 B: The pivots of a triangular matrix lie directly on its main diagonal. $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$ when all diagonal entries are nonzero.

2.2 C: Success: Subtract row 1 from rows 2 and 3 to get $0y - 2z = -2$ and $-2y + 0z = -4$. Exchange rows 2 and 3 to get U: $x + y + z = 7, -2y = -4, -2z = -2$. Pivots are $1, -2, -2$; solution is $z = 1, y = 2, x = 4$. **Failure:** Subtract row 1 from row 2 and add row 1 to row 3 to get $0y - 2z = -2$ and $0y + 2z = 10$. Adding these gives $0z = 8$ (no pivot in column 2, no solution). If 3 changes to -5, we get $0 = 0$, $z = 1$, y free, and $x = 6 - y$.

2.2.3 Problem Set 2.2: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 2.2 (1–16) 2 by 2 and 3 by 3 Elimination

1. What multiple ℓ_{21} of equation 1 should be subtracted from equation 2? $2x + 3y = 1, 10x + 9y = 11$. Write down the upper triangular system and circle the two pivots.
2. Solve the triangular system of Problem 1 by back substitution, y before x. If the right side changes to $(4, 44)$, what is the new solution?
3. What multiple of equation 1 should be subtracted from equation 2? $2x - 4y = 6, -x + 5y = 0$. Solve the triangular system. If the right side changes to $(-6, 0)$, what is the new solution?
4. What multiple ℓ of equation 1 should be subtracted from equation 2 to remove c? $ax + by = f, cx + dy = g$. What is the formula for the second pivot and y?
5. Choose a right side which gives no solution and another right side which gives infinitely many solutions for $3x + 2y = 10, 6x + 4y = ?$. What are two of those solutions?
6. Choose a coefficient b that makes $2x + by = 16, 4x + 8y = g$ singular. Then choose g that makes it solvable and find two solutions.
7. For which numbers a does elimination break down (1) permanently (2) temporarily on $ax + 3y = -3, 4x + 6y = 6$? Fix the temporary breakdown and solve.
8. For which three numbers k does elimination break down on $kx + 3y = 6, 3x + ky = -6$? Which is fixed by a row exchange? Is the number of solutions 0 or 1 or ∞?
9. What test on b_1 and b_2 decides whether $3x - 2y = b_1, 6x - 4y = b_2$ allow a solution?
10. In the xy plane, draw $x + y = 5$ and $x + 2y = 6$ and the equation $y =$ ___ from elimination. The line $5x - 4y = c$ goes through the solution if $c =$ ___.
11. (a) If (x, y, z) and (X, Y, Z) are two solutions to a linear system, what is another solution? (b) If 25 planes meet at two points, where else do they meet?
12. Reduce $2x + 3y + z = 8, 4x + 7y + 5z = 20, -2y + 2z = 0$ to upper triangular form by two row operations. Circle the pivots and solve for z, y, x.
13. Apply elimination and back substitution to solve $2x - 3y = 3, 4x - 5y + z = 7, 2x - y - 3z = 5$. List the three row operations.
14. Which number d forces a row exchange on $2x + 5y + z = 0, 4x + dy + z = 2, y - z = 3$, and what is the triangular system for that d? Which d makes this system singular?
15. Which number b leads later to a row exchange on $x + by = 0, x - 2y - z = 0, y + z = 0$? Which b leads to a missing pivot? Find a nonzero solution in that singular case.
16. (a) Construct a 3×3 system that needs two row exchanges to reach a triangular form. (b) Construct a 3×3 system that needs a row exchange to keep going, but breaks down later.

SOLUTION AT END OF PAGE (Problem Set 2.2 (Problems 1–16))

1. $\ell_{21} = 5$. Triangular system: $2x + 3y = 1, -6y = 6$. Pivots: $\boxed{2}$ and $\boxed{-6}$.
2. $y = -1, x = 2$. For $(4, 44)$, the right side is multiplied by 4, so $(x, y) = (8, -4)$.
3. $\ell_{21} = -1/2$. Triangular: $2x - 4y = 6, 3y = 3 \implies y = 1, x = 5$. For $(-6, 0)$, $(x, y) = (-5, -1)$.
4. $\ell = c/a$. Second pivot: $d - \frac{cb}{a} = \frac{ad-bc}{a}$. $y = \frac{g-(c/a)f}{d-(c/a)b} = \frac{ag-cf}{ad-bc}$.
5. No solution for right side $\neq 20$ (e.g., 10). Infinite solutions for right side $= 20$; two solutions: $(0, 5)$ and $(2, 2)$.
6. $b = 4$ makes it singular; $g = 32$ makes it solvable. Two solutions: $(8, 0)$ and $(0, 4)$.
7. (1) Permanently for $a = 2$ (parallel lines, no solution). (2) Temporarily for $a = 0$.

Exchange rows: $4x + 6y = 6, 3y = -3 \implies y = -1, x = 3$.
8. Breaks down for $k = 0$ (fixed by row exchange, 1 solution), $k = 3$ ($0 = 12$, 0 solutions), and $k = -3$ ($0 = 0$, ∞ solutions).
9. Solvable (∞ solutions) if and only if $b_2 = 2b_1$; otherwise 0 solutions.
10. $y = \mathbf{1}$, and $x = 4$. Then $c = 5(4) - 4(1) = \mathbf{16}$.
11. (a) Any affine combination, such as the midpoint $\frac{1}{2}(x + X, y + Y, z + Z)$. (b) Along the entire line through those two points.
12. Subtract $2(\text{row } 1)$ from row 2 ('y+3z=4'), then add $2(\text{row } 2)$ to row 3 ('8z=8'). Pivots: $2, 1, 8$. Solution: $z = 1, y = 1, x = 2$.
13. Subtract $2(\text{row } 1)$ from row 2, $1(\text{row } 1)$ from row 3, and $2(\text{row } 2)$ from row 3. Pivots: $2, 1, -5$. Solution: $z = 0, y = 1, x = 3$.
14. $d = 10$ forces a row exchange, giving $2x + 5y + z = 0, y - z = 3, -z = 2$. $d = 11$ gives $y - z = 2$ and $y - z = 3$ (singular, no third pivot).
15. $b = -2$ forces a row exchange. $b = -1$ gives $-y - z = 0$ and $y + z = 0$ (singular); nonzero solution: $(1, 1, -1)$.
16. (a) $z = 1, x = 2, y = 3$. (b) $y + z = 1, x + y + z = 2, y + z = 3$.

BOOK EXAMPLE PS 2.2 (17–32) Singular Cases, Tridiagonal Pivots, and Challenge Problems

17. If rows 1 and 2 are the same, how far can you get with elimination? If columns 1 and 2 are the same, which pivot is missing?

18. Construct a 3×3 example with 9 different coefficients on the left side, but rows 2 and 3 become zero in elimination. How many solutions with $\mathbf{b} = (1, 10, 100)$ and with $\mathbf{b} = (0, 0, 0)$?

19. Which number q makes $x + 4y - 2z = 1, x + 7y - 6z = 6, 3y + qz = t$ singular and which t gives it infinitely many solutions? Find the solution that has $z = 1$.

20. The system is singular if row 3 of A is a ____ of the first two rows. Find a third equation that can't be solved together with $x + y + z = 0$ and $x - 2y - z = 1$.

21. Find the pivots and the solution for both 4×4 systems $A\mathbf{x} = \mathbf{b}$ (2s on diagonal, 1s adjacent) and $K\mathbf{x} = \mathbf{b}$ (2s on diagonal, -1s adjacent) with $\mathbf{b} = (0, 0, 0, 5)$.

22. If you extend Problem 21 following the $1, 2, 1$ or $-1, 2, -1$ pattern, what is the fifth pivot? What is the nth pivot?

23. If elimination leads to $x + y = 1$ and $2y = 3$, find three possible original problems.

24. For which two numbers a will elimination fail on $A = \begin{bmatrix} a & 2 \\ a & a \end{bmatrix}$?

25. For which three numbers a will elimination fail to give three pivots for $A = \begin{bmatrix} a & 2 & 3 \\ a & a & 4 \\ a & a & a \end{bmatrix}$?

26. Look for a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with row sums $a + b = 4, c + d = 8$ and column sums $a + c = 2, b + d = s$. Solvable only if $s =$ ___. Find two such matrices.

27. Elimination gives what U and what solution to $3x = 3, 6x + 2y = 8, 9x - 2y + z = 9$?

28. Create a MATLAB command `A(2,:) = ...` to subtract 3 times row 1 from row 2.

29–32. 30: If $A(5,5) = 11$ and the last pivot is $U(5,5) = 4$, what entry $A(5,5)$ would make A singular? **31:** Row j of U is a combination of which rows of A? If $A\mathbf{x} = \mathbf{0}$, is $U\mathbf{x} = \mathbf{0}$? If A is lower triangular, what is U? **32:** For a 100×100 singular matrix A, some linear combination of the 100 rows is ___, and some linear combination of the 100 columns is ___.

SOLUTION AT END OF PAGE (Problem Set 2.2 (Problems 17–32))

17. Equal rows: row 2 becomes 0, exchange with row 3 gives two pivots, missing 3rd pivot. Equal columns: missing 2nd pivot.

18. Rows $(1, 2, 3)$, $(2, 4, 6)$, and $(3, 6, 9)$. With $\mathbf{b} = (1, 10, 100)$: 0 solutions. With $\mathbf{b} = (0, 0, 0)$: ∞ solutions.

19. Subtract row 1 from row 2 to get $3y - 4z = 5$. Thus $q = -4$ makes it singular, and $t = 5$ gives ∞ solutions. For $z = 1$: $3y = 9 \implies y = 3$, and $x = 1 - 12 + 2 = -9$, so $(x, y, z) = (-9, 3, 1)$.

20. Fill in: **linear combination**. Sum of eqs. 1 and 2 is $2x - y = 1$, so $2x - y = 2$ has no solution.

21. Both have pivots $2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}$. For A: $\mathbf{x} = (-1, 2, -3, 4)$. For K: $\mathbf{x} = (1, 2, 3, 4)$.

22. Fifth pivot is $\frac{6}{5}$. The nth pivot is $\frac{n+1}{n}$.

23. Add ℓ(row 1) to row 2 for $\ell = 0, 1, 2$: (1) $x + y = 1, 2y = 3$; (2) $x + y = 1, x + 3y = 4$; (3) $x + y = 1, 2x + 4y = 5$.

24. Pivots are a and $a - 2$, so elimination fails for $a = 0$ and $a = 2$.

25. Pivots are $a, a-2, a-4$, so elimination fails for $a = 0, 2, 4$.

26. $a+b+c+d = 4+8 = 2+s \implies s = \mathbf{10}$. Two matrices: $\begin{bmatrix} 0 & 4 \\ 2 & 6 \end{bmatrix}$ and $\begin{bmatrix} 1 & 3 \\ 1 & 7 \end{bmatrix}$.

27. $U = \text{diag}(3,2,1)$ and $\mathbf{c} = (3,2,2)^T \implies x = 1, y = 1, z = 2$.

28. `A(2,:) = A(2,:) - 3*A(1,:)`.

30–32. 30: Elimination subtracted $11 - 4 = 7$ from $A(5,5)$, so $A(5,5) = 7$ makes $U(5,5) = 0$ (singular). **31:** Combination of rows $1, \ldots, j$ of A; yes, $U\mathbf{x} = \mathbf{0}$; $U\mathbf{x} = \mathbf{c} \neq \mathbf{b}$; if A is lower triangular, U is diagonal. **32:** (a) **0** (the zero row), (b) **0** (the zero column vector).

2.3 Elimination Using Matrices [Book 2.3, pp. 56–66]

2.3.1 Derivation 3: Elementary Elimination Matrices E_{ij} and Permutation Matrices P_{ij}

GOAL

Represent each elimination step and row exchange as a matrix multiplication on the left of $A\mathbf{x} = \mathbf{b}$ and on the augmented matrix $[A \;\; \mathbf{b}]$.

FROM YOUR +2 BASE

Use the row-dot-product rule $(E\mathbf{b})_i = (\text{row } i \text{ of } E) \cdot \mathbf{b}$ from Derivation 1.

Step 1: Construct the elimination matrix E_{ij}. We want a matrix E_{ij} that keeps every component of $\mathbf{b}$ unchanged except component i, which becomes $b_i - \ell_{ij} b_j$. Start with the identity matrix I (whose kth row picks out b_k) and place $-\ell_{ij}$ in row i, column j:

$$\text{For } i = 2, j = 1: \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -\ell_{21} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \implies E_{21} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 - \ell_{21} b_1 \\ b_3 \end{bmatrix}. \tag{2.15}$$

Step 2: Apply E_{ij} to a matrix A column-by-column. Require that E acts on each column of $A = [\mathbf{a}_1 \;\; \mathbf{a}_2 \;\; \mathbf{a}_3]$ separately [Book: Eq. (3), p. 59]:

$$EA = E \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} = \begin{bmatrix} E\mathbf{a}_1 & E\mathbf{a}_2 & E\mathbf{a}_3 \end{bmatrix}. \tag{2.16}$$

Because E_{ij} subtracts ℓ_{ij} times entry j from entry i in every column, $E_{ij}A$ subtracts ℓ_{ij} times row j from row i of A. Including $\mathbf{b}$ as an extra column gives the **augmented matrix** operation:

$$E \begin{bmatrix} A & \mathbf{b} \end{bmatrix} = \begin{bmatrix} EA & E\mathbf{b} \end{bmatrix}. \tag{2.17}$$

Step 3: Construct the row exchange matrix P_{ij}. To exchange row i and row j of any matrix A, exchange row i and row j of the identity matrix I to form P_{ij}:

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \implies P_{23} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_3 \\ b_2 \end{bmatrix}. \tag{2.18}$$

RESULTAssociative vs. Commutative Law [Book p. 59]

Matrix multiplication is **associative** but **not commutative**:

$$\boxed{A(BC) = (AB)C} \quad \text{(always true)}, \qquad \boxed{AB \neq BA} \quad \text{(usually different)}. \tag{2.19}$$

Multiplying by E on the **left** (EA) acts on the **rows** of A. Multiplying by E on the **right** (AE) acts on the **columns** of A.

2.3.2 Verbatim Book Examples and Worked Examples (Section 2.3)

BOOK EXAMPLE 2.3 A Left vs. Right Multiplication by E_{21} and P_{32}

What 3 by 3 matrix E_{21} subtracts 4 times row 1 from row 2? What matrix P_{32} exchanges row 2 and row 3? If you multiply A on the *right* instead of the left, describe the results AE_{21} and AP_{32}.

BOOK EXAMPLE 2.3 B Elimination on the Augmented Matrix $[A\ \mathbf{b}]$

Write down the augmented matrix $[A\ \mathbf{b}]$ with an extra column:

$$\begin{aligned} x + 2y + 2z &= 1 \\ 4x + 8y + 9z &= 3 \\ 3y + 2z &= 1. \end{aligned}$$

Apply E_{21} and then P_{32} to reach a triangular system. Solve by back substitution. What combined matrix $P_{32}E_{21}$ will do both steps at once?

BOOK EXAMPLE 2.3 C Multiplying Matrices in Two Ways

Multiply these matrices in two ways. First, rows of A times columns of B. Second, **columns of A times rows of B**. That unusual way produces two matrices that add to AB. How many separate ordinary multiplications are needed?

$$AB = \begin{bmatrix} 3 & 4 \\ 1 & 5 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 16 \\ 7 & 9 \\ 4 & 8 \end{bmatrix}.$$

SOLUTION AT END OF PAGE (Worked Examples 2.3 A, 2.3 B, and 2.3 C)

2.3 A: Performing the operations on I gives:

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad P_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Multiplying on the right (AE_{21}) subtracts 4 times **column 2** from **column 1**. Multiplying AP_{32} exchanges **columns 2 and 3**.

2.3 B:

$$[A\ \mathbf{b}] = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 4 & 8 & 9 & 3 \\ 0 & 3 & 2 & 1 \end{bmatrix} \xrightarrow{E_{21}} \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 3 & 2 & 1 \end{bmatrix} \xrightarrow{P_{32}} \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

Back substitution gives $z = -1$, $3y - 2 = 1 \implies y = 1$, and $x + 2 - 2 = 1 \implies x = 1$. The combined matrix is $P_{32}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -4 & 1 & 0 \end{bmatrix}$.

2.3 C: Rows times columns uses 6 dot products (2 multiplications each, $3 \cdot 2 \cdot 2 = 12$ multiplications total). Columns of A times rows of B gives two 3×2 outer-product matrices that add to AB:

$$AB = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 2 & 4 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 5 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 10 & 16 \\ 7 & 9 \\ 4 & 8 \end{bmatrix}.$$

2.3.3 Problem Set 2.3: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 2.3 (1–15) Elimination Matrices and Row Exchanges

1. Write down the 3 by 3 matrices that produce these elimination steps: (a) E_{21} subtracts 5 times row 1 from row 2. (b) E_{32} subtracts -7 times row 2 from row 3. (c) P exchanges rows 1 and 2, then rows 2 and 3.

2. In Problem 1, applying E_{21} and then E_{32} to $\mathbf{b} = (1,0,0)$ gives $E_{32}E_{21}\mathbf{b} =$ ____. Applying E_{32} before E_{21} gives $E_{21}E_{32}\mathbf{b} =$ ____. When E_{32} comes first, row ___ feels no effect from row ___.

3. Which three matrices E_{21}, E_{31}, E_{32} put $A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix}$ into triangular form U? Multiply those E's to get one matrix M that does elimination: $MA = U$.

4. Include $\mathbf{b} = (1,0,0)$ as a fourth column in Problem 3 to produce $[A \;\; \mathbf{b}]$. Carry out the elimination steps on this augmented matrix to solve $A\mathbf{x} = \mathbf{b}$.

5. Suppose $a_{33} = 7$ and the third pivot is 5. If you change a_{33} to 11, the third pivot is ___. If you change a_{33} to ___, there is no third pivot.

6. If every column of A is a multiple of $(1,1,1)$, then $A\mathbf{x}$ is always a multiple of $(1,1,1)$. Do a 3×3 example. How many pivots are produced by elimination?

7. Suppose E subtracts 7 times row 1 from row 3. (a) To *invert* that step you should ___ 7 times row ___ to row ___. (b) What "inverse matrix" E^{-1} takes that reverse step? (c) Show $EE^{-1} = I$.

8. The **determinant** of $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\det M = ad - bc$. Subtract ℓ times row 1 from row 2 to produce M^*. Show that $\det M^* = \det M$ for every ℓ.

9. (a) E_{21} subtracts row 1 from row 2 and then P_{23} exchanges rows 2 and 3. What is $M = P_{23}E_{21}$? (b) P_{23} exchanges rows 2 and 3 and then E_{31} subtracts row 1 from row 3. What is $M = E_{31}P_{23}$?

10. (a) What 3×3 matrix E_{13} adds row 3 to row 1? (b) What matrix adds row 1 to row 3 and *at the same time* row 3 to row 1? (c) What matrix adds row 1 to row 3 and *then* adds row 3 to row 1?

11. Create a matrix with $a_{11} = a_{22} = a_{33} = 1$ where elimination produces two negative pivots without row exchanges.

12. Multiply: $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 4 & 0 \end{bmatrix}$.

13–15. 13: Explain why a zero third column of B stays zero in EB, but a zero third row of B might not stay zero. **14:** Find E_{21}, E_{32}, E_{43} for the 4×4 $-1, 2, -1$ matrix A. **15:** Write the 3×3 matrix with $a_{ij} = 2i - 3j$ and find E_{32}.

SOLUTION AT END OF PAGE (Problem Set 2.3 (Problems 1–15))

1. (a) $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, (b) $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$, (c) $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

2. $E_{32}E_{21}\mathbf{b} = (1, -5, -35)^T$. $E_{21}E_{32}\mathbf{b} = (1, -5, 0)^T$. Row **3** feels no effect from row **1**.

3. $\ell_{21}=4, \ell_{31}=-2, \ell_{32}=2 \implies M=E_{32}E_{31}E_{21}=\begin{bmatrix}1&0&0\\-4&1&0\\10&-2&1\end{bmatrix}$, $U=\begin{bmatrix}1&1&0\\0&2&1\\0&0&-2\end{bmatrix}$.

4. $M\mathbf{b}=(1,-4,10)^T$. Back substitution on $U\mathbf{x}=(1,-4,10)^T$ gives $x_3=-5, x_2=1/2, x_1=1/2$.

5. Increasing a_{33} by 4 increases the third pivot by 4 to **9**. Changing a_{33} to **2** gives 0 (no third pivot).

6. Only **1** pivot is produced (rows 2 and 3 become zero after step 1).

7. (a) **Add** 7 times row **1** to row **3**. (b) E^{-1} has $+7$ in the $(3,1)$ position. (c) $EE^{-1}=I$.

8. $\det M^* = a(d-\ell b)-b(c-\ell a)=ad-bc=\det M$.

9. Both give $M=\begin{bmatrix}1&0&0\\0&0&1\\-1&1&0\end{bmatrix}$.

10. (a) $E_{13}=\begin{bmatrix}1&0&1\\0&1&0\\0&0&1\end{bmatrix}$. (b) $\begin{bmatrix}1&0&1\\0&1&0\\1&0&1\end{bmatrix}$. (c) $\begin{bmatrix}1&0&1\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\1&0&1\end{bmatrix}=\begin{bmatrix}2&0&1\\0&1&0\\1&0&1\end{bmatrix}$.

11. $A=\begin{bmatrix}1&2&0\\1&1&2\\0&1&1\end{bmatrix}$ has pivots $1,-1,-3$. **12.** First product reverses rows and columns: $\begin{bmatrix}9&8&7\\6&5&4\\3&2&1\end{bmatrix}$. Second product: $\begin{bmatrix}1&2&3\\0&1&-2\\0&2&-3\end{bmatrix}$.

13–15. **13:** Column 3 of EB is $E(\text{col 3 of } B)=E\mathbf{0}=\mathbf{0}$; row 3 of EB combines all rows of B. **14:** $\ell_{21}=1/2, \ell_{32}=2/3, \ell_{43}=3/4$, so E_{21}, E_{32}, E_{43} have $+1/2, +2/3, +3/4$ in positions $(2,1),(3,2),(4,3)$. **15:** $A=\begin{bmatrix}-1&-4&-7\\1&-2&-5\\3&0&-3\end{bmatrix}$; E_{31} changes $a_{32}=0$ to -12, so E_{32} has $-\ell_{32}=-2$ in position $(3,2)$.

BOOK EXAMPLE PS 2.3 (16–31) Creating Matrices, Augmented Systems, and Challenge Problems

16. Write in 2×2 matrix form $A\mathbf{x} = \mathbf{b}$ and solve: (a) X is twice as old as Y and their ages add to 33. (b) $(x, y) = (2, 5)$ and $(3, 7)$ lie on the line $y = mx + c$. Find m and c.

17. The parabola $y = a + bx + cx^2$ goes through $(1, 4), (2, 8), (3, 14)$. Find and solve a matrix equation for (a, b, c).

18. Multiply $E = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}$ and $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix}$ in the orders EF and FE. Also compute E^2 and F^3 and F^{100}.

19. Multiply $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $Q = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ in orders PQ, QP, P^2.

20. (a) If all columns of B are the same, all columns of EB are the same because each is E times ___. (b) If all rows of B are $[1 \;\; 2 \;\; 4]$, the rows of EB are ___.

21. If E adds row 1 to row 2 and F adds row 2 to row 1, does EF equal FE?

22. If E_{21} subtracts row 1 from row 2, write a formula for: (a) 3rd component of $A\mathbf{x}$, (b) $(2, 1)$ entry of $E_{21}A$, (c) $(2, 1)$ entry of $E_{21}(E_{21}A)$, (d) 1st component of $E_{21}A\mathbf{x}$.

23. For $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$, what is the effect of $E(EA)$? In the opposite order AE, we are subtracting 2 times ___ of A from ___.

24. Apply elimination to $[A \;\; \mathbf{b}]$ for $A\mathbf{x} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 17 \end{bmatrix}$. Find $U\mathbf{x} = \mathbf{c}$ and $\mathbf{x}$.

25. Apply elimination to $[A \;\; \mathbf{b}]$ for $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}$. Change 6 so there *is* a solution.

26. Solve $\begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ at once using a 2×4 matrix.

27–31. 27: Choose a, b, c, d in $[A \;\; \mathbf{b}] = \begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 4 & 5 & b \\ 0 & 0 & d & c \end{bmatrix}$ for (a) no solution, (b) ∞ solutions. **28:** If $AB = I$ and $BC = I$, prove $A = C$. **29:** Find E and M to reduce Pascal's 4×4 matrix to I. **30:** Factor $M = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$ using $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. **31:** Multiply $E_{43}E_{32}E_{21}$ for 4×4 matrix K.

SOLUTION AT END OF PAGE (Problem Set 2.3 (Problems 16–31))

16. (a) $\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 0 \\ 33 \end{bmatrix} \implies X = 22, Y = 11$. (b) $\begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix} \implies m = 2, c = 1$.

17. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 14 \end{bmatrix} \implies a = 2, b = 1, c = 1$.

18. $EF = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$, $FE = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ ac+b & c & 1 \end{bmatrix}$, $E^2 = \begin{bmatrix} 1 & 0 & 0 \\ 2a & 1 & 0 \\ 2b & 0 & 1 \end{bmatrix}$, $F^n = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & nc & 1 \end{bmatrix}$.

19. $PQ = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $QP = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $P^2 = I$.

20. (a) **that column of** B. (b) **multiples of** $[1 \quad 2 \quad 4]$. **21.** No: $EF = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \neq FE = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

22. (a) $\sum_{j=1}^{n} a_{3j}x_j$, (b) $a_{21} - a_{11}$, (c) $a_{21} - 2a_{11}$, (d) $\sum_{j=1}^{n} a_{1j}x_j$.

23. $E(EA)$ subtracts 4(row 1) from row 2. AE subtracts 2(**column 2**) from **column 1**.

24. $U\mathbf{x} = \mathbf{c}$ is $\begin{bmatrix} 2 & 3 \\ 0 & -5 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 15 \end{bmatrix} \implies x_2 = -3, x_1 = 5$.

25. Row 3 − Row 2 − Row 1 gives $0 = 6 - 2 - 1 = 3$ (no solution). Change 6 to **3**.

26. $\begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{bmatrix} \implies \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7 \\ 2 \end{bmatrix}, \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$.

27–31. 27: (a) $d = 0, c \neq 0$; (b) $d = 0, c = 0$; a and b have no effect. **28:** $A = AI = A(BC) = (AB)C = IC = C$. **29:** E has -1s on subdiagonal; M has alternating $\pm\binom{i}{j}$. **30:** $M = ABABA$.

31: $E_{43}E_{32}E_{21} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 1/3 & 2/3 & 1 & 0 \\ 1/4 & 2/4 & 3/4 & 1 \end{bmatrix}$.

2.4 Rules for Matrix Operations [Book 2.4, pp. 67–80]

2.4.1 Derivation 4: Four Ways to Multiply AB, Block Multiplication, and Schur Complement

GOAL

Derive the four equivalent ways to multiply an $m \times n$ matrix A by an $n \times p$ matrix B, and derive the block elimination Schur complement $S = D - CA^{-1}B$.

FROM YOUR +2 BASE

Use **T3** (dot product) and the column rule $A[\mathbf{b}_1 \dots \mathbf{b}_p] = [A\mathbf{b}_1 \dots A\mathbf{b}_p]$.

Step 1: Shape compatibility and the (i, j) entry (Row-Column Rule). If A is $m \times n$ and B is $n \times p$, the product AB is $m \times p$. Column j of AB is A(column j of B). By equation (2.8), the ith entry of that column is the dot product of row i of A with column j of B:

$$\boxed{(AB)_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B) = \sum_{k=1}^{n} a_{ik} b_{kj}}. \tag{2.20}$$

Each dot product takes n multiplications. For $m \times p$ entries, AB uses $\boxed{mnp}$ multiplications.

Step 2: Sum of Rank-One Outer Products (Column-Row Rule). Split the sum over $k = 1, \dots, n$ in equation (2.20) into n separate $m \times p$ matrices [Book: Eq. (2), p. 71]:

$$\boxed{AB = \sum_{k=1}^{n} (\text{column } k \text{ of } A)(\text{row } k \text{ of } B) = \mathbf{a}_1\mathbf{b}_1^* + \mathbf{a}_2\mathbf{b}_2^* + \cdots + \mathbf{a}_n\mathbf{b}_n^*}. \tag{2.21}$$

Step 3: Block Multiplication and Block Elimination (Schur Complement). When cuts between columns of A match cuts between rows of B, we multiply blocks exactly like numbers [Book: Eq. (1), p. 70]. For a 2×2 block matrix with invertible upper-left block A, multiply the first block row by CA^{-1} and subtract from the second block row [Book: Eq. (4), p. 71]:

$$\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}. \tag{2.22}$$

The final block $\boxed{S = D - CA^{-1}B}$ is the **Schur complement** (matching $d - cb/a$ for numbers).

2.4.2 Verbatim Book Examples and Worked Examples (Section 2.4)

BOOK EXAMPLE 2.4 A Pascal Matrices and Hypercubes

Here is the lower triangular Pascal matrix $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix}$ with entries $\ell_{ij} = \binom{i}{j}$.

1. What is $H = L^2$? This is the "hypercube matrix".
2. Multiply H times **ones** $(1, 1, 1, 1)^T$ and **powers** $(1, x, x^2, x^3)^T$.
3. The last row of H is $8, 12, 6, 1$. A cube has 8 corners, 12 edges, 6 faces, 1 box. *What would the next row of H tell about a hypercube in* **4***D?*

BOOK EXAMPLE 2.4 B Commutativity and Associativity

For these matrices, when does $AB = BA$? When does $BC = CB$? When does A times BC equal AB times C? Give the conditions on their entries p, q, r, z:

$$A = \begin{bmatrix} p & 0 \\ q & r \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & z \\ 0 & 0 \end{bmatrix}.$$

If $p, q, r, 1, z$ are 4 by 4 blocks instead of numbers, do the answers change?

BOOK EXAMPLE 2.4 C Adjacency Matrices and Graph Paths

A **directed graph** has 2 nodes with edges $(1 \to 1), (1 \to 2), (2 \to 1)$, giving adjacency matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. Why does the i, j entry of A^2, which is $\sum a_{ik}a_{kj}$, count the *two-step paths* from i to j? List all of the 3-step paths between each pair of nodes and compare with A^3.

SOLUTION AT END OF PAGE (Worked Examples 2.4 A, 2.4 B, and 2.4 C)

2.4 A: 1. $H = L^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 4 & 1 & 0 \\ 8 & 12 & 6 & 1 \end{bmatrix}$. 2. $H(1,1,1,1)^T = (1,3,9,27)^T$ (powers of 3). $H(1, x, x^2, x^3)^T = (1, 2+x, (2+x)^2, (2+x)^3)^T$. 3. By the rule $2h_{ij} + h_{i,j-1} = h_{i+1,j}$, the next row is **16, 32, 24, 8, 1**: a 4D hypercube has 16 corners, 32 edges, 24 faces, 8 boxes, and 1 hypercube.

2.4 B: $AB = \begin{bmatrix} p & p \\ q & q+r \end{bmatrix}$ and $BA = \begin{bmatrix} p+q & r \\ q & r \end{bmatrix}$, so $AB = BA \iff q = 0$ and $p = r$. $BC = \begin{bmatrix} 0 & z \\ 0 & 0 \end{bmatrix} = CB$ for all z. $A(BC) = (AB)C$ is **always true** for all p, q, r, z. With 4×4 blocks (and $1 \to I$), the answers do not change.

2.4 C: The product $a_{ik}a_{kj}$ equals 1 if and only if both edges $(i \to k)$ and $(k \to j)$ exist, so $\sum_k a_{ik}a_{kj}$ counts all 2-step paths from i to j. We find $A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $A^3 = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ (Fibonacci numbers F_{k+1}, F_k, F_{k-1}). The 3-step paths are: $(1 \to 1)$: 1-1-1-1, 1-1-2-1, 1-2-1-1 (3 paths); $(1 \to 2)$: 1-1-1-2, 1-2-1-2 (2 paths); $(2 \to 1)$: 2-1-1-1, 2-1-2-1 (2 paths); $(2 \to 2)$: 2-1-1-2 (1 path).

2.4.3 Problem Set 2.4: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 2.4 (1–20) Laws of Matrix Multiplication and Powers

1. A is 3×5, B is 5×3, C is 5×1, and D is 3×1. All entries are 1. Which of these matrix operations are allowed, and what are the results? $BA, AB, ABD, DBA, A(B+C)$.

2. What rows or columns or matrices do you multiply to find: (a) 3rd column of AB? (b) 1st row of AB? (c) entry in row 3, column 4 of AB? (d) entry in row 1, column 1 of CDE?

3–4. For $A = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}$, verify $AB + AC = A(B+C)$ and $A(BC) = (AB)C$.

5. Compute A^2, A^3 and predict A^5, A^n for $A = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}$.

6. Show $(A+B)^2 \neq A^2 + 2AB + B^2$ for $A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$. Write the correct rule for $(A+B)(A+B)$.

7. True or false: (a) If columns 1 and 3 of B are the same, so are columns 1 and 3 of AB. (b) If rows 1 and 3 of B are the same, so are rows 1 and 3 of AB. (c) If rows 1 and 3 of A are the same, so are rows 1 and 3 of ABC. (d) $(AB)^2 = A^2B^2$.

8. How is each row of DA and EA related to the rows of A, and each column of AD and AE related to the columns of A, for $D = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}, E = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$?

9–10. In 9, verify $(EA)F = E(AF)$ (associative law). In 10, compare $F(EA)$ with $E(FA)$ (commutative law broken).

11. (3×3 matrices) Choose the only B so that for every matrix A: (a) $BA = 4A$, (b) $BA = 4B$, (c) BA has rows 1 and 3 of A reversed and row 2 unchanged, (d) All rows of BA are the same as row 1 of A.

12. Prove that if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ commutes with $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then $A = aI$.

13. Which are guaranteed to equal $(A-B)^2$: $A^2 - B^2$, $(B-A)^2$, $A^2 - 2AB + B^2$, $A(A-B) - B(A-B)$, $A^2 - AB - BA + B^2$?

14. True or false: (a) If A^2 is defined then A is square. (b) If AB and BA are defined then A and B are square. (c) If AB and BA are defined then AB and BA are square. (d) If $AB = B$ then $A = I$.

15. If A is $m \times n$, how many multiplications when: (a) A multiplies $\mathbf{x}$ ($n \times 1$)? (b) A multiplies B ($n \times p$)? (c) A multiplies itself ($m = n$)?

16. For $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 4 \\ 1 & 0 & 6 \end{bmatrix}$, compute only: (a) col 2 of AB, (b) row 2 of AB, (c) row 2 of A^2, (d) row 2 of A^3.

17–20. 17: Write 3×3 matrices for (a) $a_{ij} = \min(i,j)$, (b) $a_{ij} = (-1)^{i+j}$, (c) $a_{ij} = i/j$. **18:** Describe (a) $a_{ij} = 0$ if $i \neq j$, (b) $a_{ij} = 0$ if $i < j$, (c) $a_{ij} = a_{ji}$, (d) $a_{ij} = a_{1j}$. **19:** Pivots and multipliers of $A = (a_{ij})$. **20:** Compute A^2, A^3, A^4 for 4×4 shift matrix A with 2s on superdiagonal.

SOLUTION AT END OF PAGE (Problem Set 2.4 (Problems 1–20))

1. $BA = 3\,\text{ones}(5,5)$; $AB = 5\,\text{ones}(3,3)$; $ABD = 15\,\text{ones}(3,1)$; DBA and $A(B+C)$ are not allowed.

2. (a) $A(\text{col 3 of } B)$, (b) $(\text{row 1 of } A)B$, (c) $(\text{row 3 of } A)\cdot(\text{col 4 of } B)$, (d) $(\text{row 1 of } C)D(\text{col 1 of } E)$.

3–4. $AB + AC = \begin{bmatrix}0&7\\0&7\end{bmatrix} + \begin{bmatrix}3&1\\6&2\end{bmatrix} = \begin{bmatrix}3&8\\6&9\end{bmatrix} = A(B+C)$; $A(BC) = \begin{bmatrix}0&0\\0&0\end{bmatrix} = (AB)C$.

5. For $\begin{bmatrix}1&b\\0&1\end{bmatrix}$: $A^n = \begin{bmatrix}1&nb\\0&1\end{bmatrix}$. For $\begin{bmatrix}2&2\\0&0\end{bmatrix}$: $A^n = \begin{bmatrix}2^n&2^n\\0&0\end{bmatrix}$.

6. $(A+B)^2 = \begin{bmatrix}10&4\\6&6\end{bmatrix} \neq \begin{bmatrix}16&2\\3&0\end{bmatrix}$. Correct rule: $(A+B)^2 = A^2 + \mathbf{AB} + \mathbf{BA} + B^2$.

7. (a) **True**, (b) **False**, (c) **True**, (d) **False**.

8. DA multiplies row 1 by 3, row 2 by 5; EA replaces both rows with row 2. AD multiplies col 1 by 3, col 2 by 5; AE puts col 1 into col 2 and zeros col 1.

9–10. In 9, both give $\begin{bmatrix}a & a+b\\ a+c & a+b+c+d\end{bmatrix}$ (associative law). In 10, $E(FA) = \begin{bmatrix}a+c & b+d\\ a+2c & b+2d\end{bmatrix} \neq F(EA)$ (not commutative).

11. (a) $B = 4I$, (b) $B = 0$, (c) $B = \begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$, (d) $B = \begin{bmatrix}1&0&0\\1&0&0\\1&0&0\end{bmatrix}$.

12. $AB = BA \implies b = c = 0$; $AC = CA \implies a = d$, so $A = aI$.

13. $(B-A)^2$, $A(A-B) - B(A-B)$, and $A^2 - AB - BA + B^2$.

14. (a) **True**, (b) **False** (A can be $m \times n$ and B can be $n \times m$), (c) **True**, (d) **False**.

15. (a) mn, (b) mnp, (c) n^3. **16.** (a) $(0,0)^T$, (b) $[1\ \ 0\ \ 0]$, (c) $[0\ \ 1]$, (d) $[3\ \ -2]$.

17–20. 18: (a) Diagonal, (b) Lower triangular, (c) Symmetric, (d) Equal rows; 0 matrix belongs to all four. **19:** (a) a_{11}, (b) $\ell_{31} = a_{31}/a_{11}$, (c) $a_{32} - \frac{a_{31}a_{12}}{a_{11}}$, (d) $a_{22} - \frac{a_{21}a_{12}}{a_{11}}$. **20:** A^2 has 4s on 2nd superdiagonal, A^3 has 8 in $(1,4)$ corner, $A^4 = 0$.

BOOK EXAMPLE PS 2.4 (21–37) Block Multiplication, Outer Products, and Challenge Problems

21–24. 21: Powers of $A = [.5\ .5; .5\ .5]$ and AB with $B = [1\ 0; 0\ -1]$. **22:** Find real 2×2 matrices with $A^2 = -I$, $BC = 0$, $DE = -ED \neq 0$. **23:** Nonzero A with $A^2 = 0$, and $A^2 \neq 0$ but $A^3 = 0$. **24:** Predict A^n for $A_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $A_3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$.

25–26. Multiply A times I and $AB = \begin{bmatrix} 1 & 0 \\ 2 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ using **columns times rows**.

27. Prove the product of upper triangular matrices is always upper triangular.

28. State the cuts in A and B for the four multiplication rules.

29–30. For $A = \begin{bmatrix} 2 & 1 & 0 \\ -2 & 0 & 1 \\ 8 & 5 & 3 \end{bmatrix}$, find $E = E_{31}E_{21}$, $\mathbf{c}$, D, and the Schur complement $D - \mathbf{cb}/a$.

31. Use blocks to separate $(A + iB)(\mathbf{x} + i\mathbf{y}) = (A\mathbf{x} - B\mathbf{y}) + i(B\mathbf{x} + A\mathbf{y})$.

32–33. If $A\mathbf{x}_1 = \mathbf{e}_1, A\mathbf{x}_2 = \mathbf{e}_2, A\mathbf{x}_3 = \mathbf{e}_3$, what is AX for $X = [\mathbf{x}_1\ \ \mathbf{x}_2\ \ \mathbf{x}_3]$? If $\mathbf{x}_1 = (1,1,1), \mathbf{x}_2 = (0,1,1), \mathbf{x}_3 = (0,0,1)$, solve $A\mathbf{x} = (3,5,8)$ and find A.

34–37. 34: Find all $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ that commute with $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. **35:** Adjacency matrix A and A^2 for a 4-node circle graph. **36:** Multiplication count for $(AB)C$ vs. $A(BC)$ and $(\mathbf{u}^T\mathbf{v})\mathbf{w}^T$ vs. $\mathbf{u}^T(\mathbf{v}\mathbf{w}^T)$. **37:** Prove $(AB)C = A(BC)$ and show left-inverse B equals right-inverse C.

SOLUTION AT END OF PAGE (Problem Set 2.4 (Problems 21–37))

21–24. 21: $A^n = A$; $(AB)^2 = 0$. **22:** $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $E = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. **23:** $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $(A^2 = 0)$; $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $(A^3 = 0)$. **24:** $A_1^n = \begin{bmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{bmatrix}$, $A_2^n = 2^{n-1}A_2$, $A_3^n = a^{n-1}A_3$.

25–26. $AB = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 3 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 0 \\ 6 & 6 & 0 \\ 6 & 6 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 4 & 8 & 4 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 0 \\ 10 & 14 & 4 \\ 7 & 8 & 1 \end{bmatrix}$.

27. For $i > j$, (row i of A) $\cdot$ (col j of B) $= \sum_{k=1}^{n} a_{ik}b_{kj} = 0$ because $a_{ik} = 0$ for $k < i$ and $b_{kj} = 0$ for $k > j$.

28. (1) No cuts in A, vertical cuts in B; (2) Horizontal cuts in A, no cuts in B; (3) Horizontal in A, vertical in B; (4) Vertical in A, horizontal in B.

29–30. $E = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} -2 \\ 8 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 1 \\ 5 & 3 \end{bmatrix}$, and $D - \frac{\mathbf{cb}}{a} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$.

31. $\begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} A\mathbf{x} - B\mathbf{y} \\ B\mathbf{x} + A\mathbf{y} \end{bmatrix}$.

32–33. $AX = I$ (so $X = A^{-1}$). For $\mathbf{b} = (3,5,8)$, $\mathbf{x} = 3\mathbf{x}_1 + 5\mathbf{x}_2 + 8\mathbf{x}_3 = (3,8,16)^T$, and $A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$.

34–37. 34: $a+b=a+c$ and $c+d=b+d \implies b=c$ and $a=d$. **35:** A has 0 on diagonal and 1s on adjacent cyclic diagonals; $A^2 = 2\begin{bmatrix}1&0&1&0\\0&1&0&1\\1&0&1&0\\0&1&0&1\end{bmatrix}$. **36:** (a) $(AB)C$ uses $2(4)(7)+2(7)(10) = 196 < 280+560 = 840$, so prefer $(AB)C$. (b) $(\mathbf{u}^T\mathbf{v})\mathbf{w}^T$ uses $2N$ vs. $2N^2$, so prefer $(\mathbf{u}^T\mathbf{v})\mathbf{w}^T$. **37:** Fill in: **columns**; $B = BI = B(AC) = (BA)C = IC = C$.

2.5 Inverse Matrices [Book 2.5, pp. 81–94]

2.5.1 Derivation 5: Inverse Matrix A^{-1}, Reverse-Order Product Rule, and Gauss-Jordan Method

GOAL

Define the two-sided inverse A^{-1}, derive the 2×2 inverse formula, prove $(AB)^{-1} = B^{-1}A^{-1}$, and derive the Gauss-Jordan elimination algorithm $[A \;\; I] \to [I \;\; A^{-1}]$.

FROM YOUR +2 BASE

Use **T1** (solving 2×2 equations) and the associative law $A(BC) = (AB)C$.

Step 1: Uniqueness of Left and Right Inverses. A square matrix A is **invertible** if there exists a matrix A^{-1} such that [Book: Eq. (1), p. 81]:

$$\boxed{A^{-1}A = I \quad \text{and} \quad AA^{-1} = I}. \tag{2.23}$$

If $BA = I$ (left-inverse) and $AC = I$ (right-inverse), the associative law proves $B = C$ [Book: Eq. (2), p. 81]:

$$B(AC) = (BA)C \implies BI = IC \implies B = C. \tag{2.24}$$

When A is invertible, $A\mathbf{x} = \mathbf{b}$ has the unique solution $\boxed{\mathbf{x} = A^{-1}\mathbf{b}}$. If $A\mathbf{x} = \mathbf{0}$ for any nonzero vector $\mathbf{x} \neq \mathbf{0}$, then A **cannot** have an inverse.

Step 2: Inverse of a Product AB and ABC. Multiply AB by $B^{-1}A^{-1}$ and use the associative law to group $BB^{-1} = I$ in the middle [Book: Eqs. (4)–(5), p. 82]:

$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I \\ \implies \boxed{(AB)^{-1} = B^{-1}A^{-1}} \quad &\text{and} \quad \boxed{(ABC)^{-1} = C^{-1}B^{-1}A^{-1}}.\end{aligned} \tag{2.25}$$

Step 3: Gauss-Jordan Elimination $[A \;\; I] \to [I \;\; A^{-1}]$. Write $AA^{-1} = I$ column-by-column, where $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are the columns of A^{-1} and $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the columns of I [Book: Eq. (7), p. 83]:

$$AA^{-1} = A\begin{bmatrix}\mathbf{x}_1 & \mathbf{x}_2 & \ldots & \mathbf{x}_n\end{bmatrix} = \begin{bmatrix}\mathbf{e}_1 & \mathbf{e}_2 & \ldots & \mathbf{e}_n\end{bmatrix} = I. \tag{2.26}$$

Solve all n systems simultaneously by placing I next to A in the $n \times 2n$ block matrix $[A \;\; I]$. Row reduction (downward below pivots, upward above pivots, and dividing by pivots) multiplies the entire block matrix on the left by A^{-1}:

$$\boxed{A^{-1}\begin{bmatrix}A & I\end{bmatrix} = \begin{bmatrix}I & A^{-1}\end{bmatrix}}. \tag{2.27}$$

For $n = 2$, reducing $\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right]$ gives the **2 by 2 inverse formula** [Book: Eq. (3), p. 81]:

$$\boxed{\begin{bmatrix}a & b \\ c & d\end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix}d & -b \\ -c & a\end{bmatrix}} \qquad (\text{invertible if and only if } ad - bc \neq 0). \tag{2.28}$$

2.5.2 Verbatim Book Examples and Worked Examples (Section 2.5)

BOOK EXAMPLE 2.5-GJ Gauss-Jordan on the $-1, 2, -1$ Matrix K

Invert the 3 by 3 tridiagonal matrix $K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by Gauss-Jordan elimination on $[K \;\; I]$.

BOOK EXAMPLE 2.5 A Inverse of a Triangular Difference Matrix

The inverse of a triangular **difference matrix** A is a triangular **sum matrix** S:

$$[A \;\; I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array}\right] = [I \;\; A^{-1}].$$

If I change a_{13} to -1, why can't this new A be inverted?

BOOK EXAMPLE 2.5 B Invertible vs. Singular Matrices

Three of these matrices are invertible, and three are singular. Find the inverse when it exists. Give reasons for noninvertibility for the other three. The matrices are in the order A, B, C, D, S, E:

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 \\ 8 & 7 \end{bmatrix} \quad \begin{bmatrix} 6 & 6 \\ 6 & 0 \end{bmatrix} \quad \begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

BOOK EXAMPLE 2.5 C Gauss-Jordan on the Triangular Pascal Matrix L

Apply the Gauss-Jordan method to invert the 4×4 triangular "Pascal matrix" $L =$ abs(pascal(4,1)).

SOLUTION AT END OF PAGE (Examples 2.5-GJ, 2.5 A, 2.5 B, and 2.5 C)

2.5-GJ: Forward elimination on $[K \;\; I]$ gives pivots $2, \frac{3}{2}, \frac{4}{3}$ (product $= 4 = \det K$). Backward elimination and dividing by the pivots yields:

$$K^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

2.5 A: If $a_{13} = -1$, all rows of A add to zero, so $A\mathbf{x} = \mathbf{0}$ has the nonzero solution $\mathbf{x} = (1, 1, 1)^T$. By Note 4, A cannot be inverted.

2.5 B: Invertible matrices:

$$B^{-1} = \frac{1}{4}\begin{bmatrix} 7 & -3 \\ -8 & 4 \end{bmatrix}, \qquad C^{-1} = \frac{1}{36}\begin{bmatrix} 0 & 6 \\ 6 & -6 \end{bmatrix}, \qquad S^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

A is singular because $\det A = 24 - 24 = 0$; D is singular because row 2 equals row 1 (only one pivot); E is singular because column 1 equals column 2 ($E(-1, 1, 0)^T = \mathbf{0}$).

2.5 C: Row reduction on $[L \quad I]$ subtracts row 1 from rows 2, 3, 4, then multiples 2, 3 of row 2 from rows 3, 4, and 3(row 3) from row 4, giving L^{-1} with alternating signs on the diagonals:

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}.$$

2.5.3 Problem Set 2.5: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 2.5 (1–21) Inverses, Singular Tests, and Properties

1. Find the inverses of $A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix}$, and $C = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$.

2. Find P^{-1} for $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

3. Solve for the first column (x, y) and second column (t, z) of A^{-1} for $A = \begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix}$.

4. Show $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ is not invertible by trying to solve $AA^{-1} = I$ for column 1.

5. Find an upper triangular U (not diagonal) with $U^2 = I$ which gives $U = U^{-1}$.

6. (a) If A is invertible and $AB = AC$, prove $B = C$. (b) If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, find $B \neq C$ with $AB = AC$.

7. If A has row 1 + row 2 = row 3, show A is not invertible: (a) Why can't $A\mathbf{x} = (1, 0, 0)$ have a solution? (b) Which (b_1, b_2, b_3) allow a solution? (c) What happens to row 3 in elimination?

8. If A has column 1 + column 2 = column 3, show A is not invertible: (a) Find a nonzero solution to $A\mathbf{x} = \mathbf{0}$. (b) Explain why there is no third pivot.

9. If A is invertible and you exchange its first two rows to reach B, how do you find B^{-1} from A^{-1}?

10. Find the inverses of $A = \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 5 \\ 0 & 0 & 7 & 6 \end{bmatrix}$.

11. (a) Find invertible A, B such that $A + B$ is singular. (b) Find singular A, B such that $A + B$ is invertible.

12–13. If $C = AB$ is invertible, find A^{-1} in terms of C^{-1} and B. If $M = ABC$ is invertible, find B^{-1}.

14. The inverse of $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ A is _____. **15.** Prove a matrix with a column of zeros has no inverse.

16. Multiply $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. **17.** Find E and $L = E^{-1}$ for subtracting row 1 from rows 2, 3 then row 2 from row 3. **18.** If $B = (A^2)^{-1}$, show $AB = A^{-1}$.

19–21. **19:** Find a, b in $(5I_4 - \text{ones}(4,4))^{-1}$ and $(6I_5 - \text{ones}(5,5))^{-1}$. **20:** Show $A = 4I_4 - \text{ones}(4,4)$ is singular. **21:** How many of the sixteen 2×2 $(0,1)$-matrices are invertible?

SOLUTION AT END OF PAGE (Problem Set 2.5 (Problems 1–21))

1. $A^{-1} = \begin{bmatrix} 0 & 1/4 \\ 1/3 & 0 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1/2 & 0 \\ -1 & 1/2 \end{bmatrix}$, $C^{-1} = \begin{bmatrix} 7 & -4 \\ -5 & 3 \end{bmatrix}$.

2. First $P^{-1} = P$; second $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

3. $(x, y) = (1/2, -1/5)$, $(t, z) = (-1/5, 1/10)$. **4.** $x + 2y = 1, 3x + 6y = 0 \implies 0 = -3$ (impossible). No, if $AA^{-1} = I$ fails for col 2, A is singular so col 1 would only be solvable for special right sides, not all.

5. $U = \begin{bmatrix} 1 & a \\ 0 & -1 \end{bmatrix}$ for any $a \neq 0$. **6.** (a) Multiply by A^{-1}: $A^{-1}AB = A^{-1}AC \implies B = C$. (b) $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

7. (a) Equation 1 + Equation 2 = Equation 3 gives $1 + 0 = 0$ (impossible). (b) $b_1 + b_2 = b_3$. (c) Row 3 becomes all zeros.

8. (a) $\mathbf{x} = (1, 1, -1)^T$. (b) Column 3 in U is also col 1 + col 2, so its $(3,3)$ entry is $0 + 0 = 0$.

9. $B = PA \implies B^{-1} = A^{-1}P^{-1} = A^{-1}P$ (exchange the first two **columns** of A^{-1}).

10. $A^{-1} = \begin{bmatrix} 0 & 0 & 0 & 1/5 \\ 0 & 0 & 1/4 & 0 \\ 0 & 1/3 & 0 & 0 \\ 1/2 & 0 & 0 & 0 \end{bmatrix}$, B^{-1} is block diagonal with blocks $\begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}$ and $\begin{bmatrix} 6 & -5 \\ -7 & 6 \end{bmatrix}$.

11. (a) $A = I, B = -I$. (b) $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.

12–14. 12: $A^{-1} = BC^{-1}$. **13:** $B^{-1} = CM^{-1}A$. **14:** $B^{-1} = A^{-1}\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$.

15–21. 15: If col j of A is $\mathbf{0}$, col j of BA is $B\mathbf{0} = \mathbf{0} \neq \mathbf{e}_j$. **16:** Product is $(ad - bc)I$. **17:** $E = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$. **18:** $A(AB) = A^2B = I$. **19:** $a = 2/5, b = 1/5$ (and $a = 1/3, b = 1/6$ for $n = 5$). **20:** $A(1, 1, 1, 1)^T = \mathbf{0}$. **21: 6** invertible matrices (3 with det $= 1$ and 3 with det $= -1$).

BOOK EXAMPLE PS 2.5 (22–46) Gauss-Jordan Calculations and Challenge Problems

22. Invert by Gauss-Jordan: $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 4 \\ 3 & 9 \end{bmatrix}$.

23. Reduce $[A \;\; I]$ to $[I \;\; A^{-1}]$ for $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$.

24. Use Gauss-Jordan on $[U \;\; I]$ to find U^{-1} for $U = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$.

25. Find A^{-1} and B^{-1} (if they exist) for $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$.

26. Find E_{21}, E_{12}, D^{-1} that reduce $A = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$ to I, and multiply $D^{-1}E_{12}E_{21} = A^{-1}$.

27–28. Invert $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, and $A = \begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}$.

29. True or false: (a) A 4×4 matrix with a row of zeros is not invertible. (b) Every matrix with 1's down the main diagonal is invertible. (c) If A is invertible then A^{-1} and A^2 are invertible.

30–31. **30:** For which three numbers c is $A = \begin{bmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{bmatrix}$ not invertible? **31:** Prove $A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$ is invertible if $a \neq 0$ and $a \neq b$.

32–46. **32:** Invert $A = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. **33:** Show $(P - Q)(1, 1, \ldots, 1)^T = \mathbf{0}$ for permutation matrices. **34:** Invert block matrices $\begin{bmatrix} I & 0 \\ C & I \end{bmatrix}$, $\begin{bmatrix} A & 0 \\ C & D \end{bmatrix}$, $\begin{bmatrix} 0 & I \\ I & D \end{bmatrix}$. **40:** Invert 4×4 bidiagonal matrix with 1s on diagonal and $-a, -b, -c$ on superdiagonal. **42:** Verify Woodbury-Morrison formula $(I - UV)^{-1} = I_n + U(I_m - VU)^{-1}V$. **43–45:** Second difference matrix T with $T_{11} = 1$, cyclic C, free-ends F, and block Schur complement $S = D - CA^{-1}B$.

SOLUTION AT END OF PAGE (Problem Set 2.5 (Problems 22–46))

22. $\begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$ and $\frac{1}{3}\begin{bmatrix} -9 & 4 \\ 3 & -1 \end{bmatrix}$. **23.** $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{bmatrix}$.

24. $U^{-1} = \begin{bmatrix} 1 & -a & ac - b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}$. **25.** $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$; B is singular $(B(1, 1, 1)^T = \mathbf{0})$.

26. $E_{21} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, E_{12} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, D^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \implies A^{-1} = \begin{bmatrix} 3 & -1 \\ -1 & 1/2 \end{bmatrix}$.

27–28. $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, and $\begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 0 \end{bmatrix}$.

29. (a) **True**, (b) **False** (e.g., all-ones matrix), (c) **True** ($(A^{-1})^{-1} = A$ and $(A^2)^{-1} = (A^{-1})^2$).

30–31. 30: Singular for $c = 0$ (col 2, 3 zero), $c = 2$ (rows 1, 2 equal), and $c = 7$ (cols 2, 3 equal). **31:** Pivots are $a, a-b, a-b$, which are all nonzero when $a \neq 0$ and $a \neq b$.

32–45. 32: A^{-1} has 1s on the main diagonal and 1s on the first superdiagonal, 0s elsewhere; $\mathbf{x} = (2,2,2,1)^T$. **34:** $\begin{bmatrix} I & 0 \\ -C & I \end{bmatrix}$, $\begin{bmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{bmatrix}$, $\begin{bmatrix} -D & I \\ I & 0 \end{bmatrix}$. **40:** $A^{-1} = \begin{bmatrix} 1 & a & ab & abc \\ 0 & 1 & b & bc \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}$. **43:** T has all pivots equal to 1; $T^{-1}_{ij} = n + 1 - \max(i,j)$. **44:** $C(1,1,1,1)^T = \mathbf{0}$ and $F(1,1,1,1)^T = \mathbf{0}$, so neither is invertible. **45:** For [2 3 3;4 1 0;4 0 1], $S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 3 \end{bmatrix} = \begin{bmatrix} -5 & -6 \\ -6 & -5 \end{bmatrix}$.

2.6 Elimination = Factorization: $A = LU$ [Book 2.6, pp. 95–106]

2.6.1 Derivation 6: $A = LU$, $A = LDU$, Two-Step Solve, and Operation Count $\frac{1}{3}n^3$

GOAL

Show that Gaussian elimination (without row exchanges) factors A into lower triangular L times upper triangular U, where the multipliers ℓ_{ij} go directly into L without mix-up, and derive the $\frac{1}{3}n^3$ operation count.

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Use the inverse product rule $(E_{32}E_{31}E_{21})^{-1} = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$ and **T4** ($\sum_{k=1}^{n} k^2 \approx \frac{1}{3}n^3$).

Step 1: Invert the elimination matrices in reverse order. For a 3×3 matrix A requiring no row exchanges, forward elimination reaches U via [Book: Eq. (1), p. 95]:

$$(E_{32}E_{31}E_{21})A = U \implies A = (E_{21}^{-1}E_{31}^{-1}E_{32}^{-1})U = LU. \tag{2.29}$$

Step 2: Prove why the multipliers ℓ_{ij} sit directly in L without mix-up. When elimination computes row 3 of U, it subtracts multiples of **earlier rows of** U (the settled pivot rows) from row 3 of A [Book: Eqs. (2)–(3), p. 97]:

$$\begin{aligned} &\text{Row 3 of } U = (\text{Row 3 of } A) - \ell_{31}(\text{Row 1 of } U) - \ell_{32}(\text{Row 2 of } U) \\ \implies &\text{Row 3 of } A = \ell_{31}(\text{Row 1 of } U) + \ell_{32}(\text{Row 2 of } U) + 1(\text{Row 3 of } U) = \begin{bmatrix} \ell_{31} & \ell_{32} & 1 \end{bmatrix} U. \end{aligned} \tag{2.30}$$

Stacking all rows $i = 1, \ldots, n$ proves that L has 1's on the diagonal and the exact multipliers ℓ_{ij} below the diagonal:

$$\boxed{L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}}. \tag{2.31}$$

Dividing each row of U by its pivot d_i factors U into $D = \operatorname{diag}(d_1, \ldots, d_n)$ and a unit upper triangular matrix. This yields the balanced factorization $\boxed{A = LDU}$.

Step 3: Solve $A\mathbf{x} = \mathbf{b}$ in two triangular steps and count operations. Substitute $A = LU$ into $A\mathbf{x} = \mathbf{b}$ to get $L(U\mathbf{x}) = \mathbf{b}$. Define $\mathbf{c} = U\mathbf{x}$ [Book: Eq. (5), p. 98]:

$$\boxed{\textbf{Step 1 (Forward):}\ L\mathbf{c} = \mathbf{b}, \qquad \textbf{Step 2 (Backward):}\ U\mathbf{x} = \mathbf{c}}. \tag{2.32}$$

By **T4**, eliminating column $1, 2, \ldots, n$ of A takes $\sum_{k=1}^{n}(k^2 - k) = \frac{1}{3}(n^3 - n) \approx \boxed{\frac{1}{3}n^3}$ multiplications and subtractions. Solving $L\mathbf{c} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{c}$ on the right side takes:

$$[(n-1) + (n-2) + \cdots + 1] + [1 + 2 + \cdots + n] = \frac{(n-1)n}{2} + \frac{n(n+1)}{2} = \boxed{n^2} \quad \text{operations.} \tag{2.33}$$

For a **band matrix** with half-bandwidth w, the Factor cost drops from $\frac{1}{3}n^3$ to $\boxed{nw^2}$, and the Solve cost drops from n^2 to $\boxed{2nw}$.

2.6.2 Verbatim Book Examples and Worked Examples (Section 2.6)

BOOK EXAMPLE 2.6 A LU Factorization of the Symmetric Pascal Matrix

Establish the amazing lower-upper factorization $P = LU$ for the 4×4 symmetric Pascal matrix:

$$\text{pascal}(4) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = LU.$$

Then predict and check the next row and column for 5 by 5 Pascal matrices.

BOOK EXAMPLE 2.6 B Two Triangular Systems for $Px = b$

The problem is: *Solve* $P\mathbf{x} = \mathbf{b} = (1,0,0,0)$ using the Pascal factors L and U:

Two triangular systems: $L\mathbf{c} = \mathbf{b}$ (forward), $U\mathbf{x} = \mathbf{c}$ (back).

SOLUTION AT END OF PAGE (Worked Examples 2.6 A and 2.6 B)

2.6 A: Subtracting row 1 of P from rows 2, 3, 4 uses multipliers $\ell_{21} = \ell_{31} = \ell_{41} = 1$. Subtracting the new row 2 from rows 3, 4 uses $\ell_{32} = 2, \ell_{42} = 3$. Subtracting 3(row 3) from row 4 uses $\ell_{43} = 3$ and leaves $U = L^T$ with all pivots equal to 1. For 5×5, row 5 of P is $[1\ \ 5\ \ 15\ \ 35\ \ 70]$, row 5 of L is $[1\ \ 4\ \ 6\ \ 4\ \ 1]$, and column 5 of U is $(1,4,6,4,1)^T$.

2.6 B: Forward substitution on $L\mathbf{c} = (1,0,0,0)^T$:

$$c_1 = 1, \qquad c_1 + c_2 = 0 \implies c_2 = -1,$$
$$c_1 + 2c_2 + c_3 = 0 \implies c_3 = 1, \qquad c_1 + 3c_2 + 3c_3 + c_4 = 0 \implies c_4 = -1.$$

Back substitution on $U\mathbf{x} = (1,-1,1,-1)^T$ from bottom to top:

$$x_4 = -1, \qquad x_3 = 1 - 3(-1) = 4,$$
$$x_2 = -1 - 2(4) - 3(-1) = -6, \qquad x_1 = 1 - (-6) - 4 - (-1) = 4.$$

Thus $\mathbf{x} = (4,-6,4,-1)^T$, which is the first column of P^{-1}.

2.6.3 Problem Set 2.6: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 2.6 (1–26) Computing $A = LU$, $A = LDU$, and Challenge Problems

1–2. Forward elimination changes $\begin{bmatrix}1&1\\1&2\end{bmatrix}\mathbf{x} = \begin{bmatrix}5\\7\end{bmatrix}$ to $\begin{bmatrix}1&1\\0&1\end{bmatrix}\mathbf{x} = \begin{bmatrix}5\\2\end{bmatrix}$. Find ℓ_{21}, L, $\mathbf{c}$, and $\mathbf{x}$.

3–4. Forward elimination on $x+y+z=5, x+2y+3z=7, x+3y+6z=11$ gives $x+y+z=5, y+2z=2, z=2$. Find ℓ_{31}, ℓ_{32}, L, U, $\mathbf{c}$, and $\mathbf{x}$.

5–7. Factor into $A = LU$: (5) $A = \begin{bmatrix}2&1&0\\0&4&2\\6&3&5\end{bmatrix}$, (6) $A = \begin{bmatrix}1&1&1\\2&4&5\\0&4&0\end{bmatrix}$, (7) $A = \begin{bmatrix}1&0&1\\2&2&2\\3&4&5\end{bmatrix}$.

8. For $A = L = \begin{bmatrix}1&0&0\\a&1&0\\b&c&1\end{bmatrix}$, multiply $E = E_{32}E_{31}E_{21}$ so $EA = I$, and multiply $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1} = L$.

9. Show directly why $\begin{bmatrix}0&1\\2&3\end{bmatrix} = LU$ and $\begin{bmatrix}1&1&0\\1&1&2\\1&2&1\end{bmatrix} = LU$ are impossible with nonzero pivots.

10. Which c in $A = \begin{bmatrix}1&c&0\\2&4&1\\3&5&1\end{bmatrix}$ leads to zero in the 2nd pivot position? Which c gives zero in the 3rd pivot position?

11–12. Find L, D, U for $A = \begin{bmatrix}2&4&8\\0&3&9\\0&0&7\end{bmatrix}$, $A = \begin{bmatrix}2&4\\4&11\end{bmatrix}$, and $B = \begin{bmatrix}1&4&0\\4&12&4\\0&4&0\end{bmatrix}$.

13–14. Compute L and U for $A = \begin{bmatrix}a&a&a&a\\a&b&b&b\\a&b&c&c\\a&b&c&d\end{bmatrix}$ and $A = \begin{bmatrix}a&r&r&r\\a&b&s&s\\a&b&c&t\\a&b&c&d\end{bmatrix}$. Find the four conditions for four pivots.

15–16. Solve $L\mathbf{c} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{c}$ for: (15) $L = \begin{bmatrix}1&0\\4&1\end{bmatrix}, U = \begin{bmatrix}2&4\\0&1\end{bmatrix}, \mathbf{b} = \begin{bmatrix}2\\11\end{bmatrix}$; (16) $L = \begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}, U = \begin{bmatrix}1&1&1\\0&1&1\\0&0&1\end{bmatrix}, \mathbf{b} = \begin{bmatrix}4\\5\\6\end{bmatrix}$.

17–26. 18: Prove LDU uniqueness from $L_1^{-1}LD = D_1U_1U^{-1}$. **19–20:** Factor tridiagonal matrices into LU and LDL^T. **22:** Eliminate upwards on $A = \begin{bmatrix}5&3&1\\3&3&1\\1&1&1\end{bmatrix}$ to find $A = UL$. **23:** If A has pivots $5, 9, 3$, what are the pivots of the upper left 2×2 submatrix A_2? **24:** Explain why $A = LU$ requires all upper-left A_k to be invertible.

SOLUTION AT END OF PAGE (Problem Set 2.6 (Problems 1–26))

1–2. $\ell_{21} = 1$, $L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $\mathbf{c} = (5,2)^T$, and $\mathbf{x} = (3,2)^T$.

3–4. $\ell_{31} = 1, \ell_{32} = 2$, $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$, $\mathbf{c} = (5,2,2)^T$, $\mathbf{x} = (5,-2,2)^T$.

5–7. (5) $L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 5 \end{bmatrix}$. (6) $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -6 \end{bmatrix}$. (7) $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

8. $E = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac-b & -c & 1 \end{bmatrix}$, while $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1} = L$ has no ac mix-up.

9–10. 9: In 2×2, $1\cdot d = 0 \implies d = 0$; in 3×3, $d = 1, e = 1, \ell = 1 \implies \ell e + f = 1 \implies f = 0$. **10:** $c = 2$ gives zero 2nd pivot; $c = 1$ gives zero 3rd pivot.

11–12. (11) $L = I, D = \text{diag}(2,3,7), U_{\text{new}} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$. (12) For A: $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, D = \text{diag}(2,3), U = L^T$. For B: $L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, D = \text{diag}(1,-4,4), U = L^T$.

13–14. Both have $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$. In 13, pivots are $a, b-a, c-b, d-c \neq 0$. In 14, pivots are $a, b-r, c-s, d-t \neq 0$.

15–16. (15) $\mathbf{c} = (2,3)^T, \mathbf{x} = (-5,3)^T$. (16) $\mathbf{c} = (4,1,1)^T, \mathbf{x} = (3,0,1)^T$; $A = LU = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$.

18–24. 18: Left side $L_1^{-1}LD$ is lower triangular, right side $D_1U_1U^{-1}$ is upper triangular, so both are diagonal $\implies L = L_1, U = U_1, D = D_1$. **22:** $A = UL = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$. **23:** 5 and 9. **24:** $A_k = L_kU_k$, so A_k is invertible iff U_k has k nonzero pivots.

2.7 Transposes and Permutations [Book 2.7, pp. 107–119]

2.7.1 Derivation 7: Transpose Rules, Inner Products, Symmetric LDL^T, and $PA = LU$

GOAL

Derive the reverse-order transpose rules $(AB)^T = B^T A^T$ and $(A^{-1})^T = (A^T)^{-1}$, the inner product identity $(A\mathbf{x})^T\mathbf{y} = \mathbf{x}^T(A^T\mathbf{y})$, the symmetric factorization $A = LDL^T$, and $PA = LU$.

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Use **T3** ($\mathbf{x}\cdot\mathbf{y} = \mathbf{x}^T\mathbf{y}$) and **T7** (integration by parts for continuous functions).

Step 1: Define A^T and prove $(AB)^T = B^T A^T$ and $(A^{-1})^T = (A^T)^{-1}$. Flipping an $m \times n$ matrix A across its main diagonal gives the $n \times m$ **transpose** A^T with entries $\boxed{(A^T)_{ij} = A_{ji}}$. For a vector $\mathbf{x}$, $A\mathbf{x}$ combines the columns of A, while the row vector $\mathbf{x}^T A^T$ combines the exact same vectors (now rows of A^T). Thus $(A\mathbf{x})^T = \mathbf{x}^T A^T$. Applying this to each column of $B = [\mathbf{x}_1 \;\; \mathbf{x}_2 \;\; \dots]$ gives [Book: Eqs. (2)–(5), pp. 107–108]:

$$(AB)^T = \begin{bmatrix} A\mathbf{x}_1 & A\mathbf{x}_2 & \dots \end{bmatrix}^T = \begin{bmatrix} \mathbf{x}_1^T A^T \\ \mathbf{x}_2^T A^T \\ \vdots \end{bmatrix} = \boxed{B^T A^T}. \tag{2.34}$$

Transposing $A^{-1}A = I$ gives $A^T(A^{-1})^T = I^T = I$, which proves $\boxed{(A^{-1})^T = (A^T)^{-1}}$.

Step 2: Inner-product definition of the transpose. The dot product of $\mathbf{x}$ and $\mathbf{y}$ is $\mathbf{x}\cdot\mathbf{y} = \mathbf{x}^T\mathbf{y}$. For any matrix A, the dot product of $A\mathbf{x}$ with $\mathbf{y}$ is:

$$\boxed{(A\mathbf{x})\cdot\mathbf{y} = (A\mathbf{x})^T\mathbf{y} = \mathbf{x}^T A^T\mathbf{y} = \mathbf{x}^T(A^T\mathbf{y}) = \mathbf{x}\cdot(A^T\mathbf{y})}. \tag{2.35}$$

For functions $x(t), y(t)$ with inner product $(x, y) = \int_{-\infty}^{\infty} x(t)y(t)\,\mathrm{d}t$, **T7** (integration by parts) gives [Book: Eq. (6), p. 109]:

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\mathrm{d}t} y(t)\,\mathrm{d}t = \int_{-\infty}^{\infty} x(t)\left(-\frac{\mathrm{d}y}{\mathrm{d}t}\right)\mathrm{d}t \implies \boxed{\left(\frac{\mathrm{d}}{\mathrm{d}t}\right)^T = -\frac{\mathrm{d}}{\mathrm{d}t}} \quad \text{(anti-symmetric)}. \tag{2.36}$$

Step 3: Symmetric matrices ($A^T = A$) and $A = LDL^T$. For any rectangular matrix R, the products $R^T R$ and RR^T are automatically symmetric because $(R^T R)^T = R^T(R^T)^T = R^T R$ [Book: Eq. (7), p. 110]. If a symmetric matrix $A = A^T$ is factored into LDU without row exchanges, then $A = A^T \implies LDU = U^T D^T L^T = U^T D L^T$. By uniqueness of LDU, $U = L^T$, so:

$$\boxed{A = A^T \implies A = LDL^T}. \tag{2.37}$$

Step 4: Permutation matrices ($P^T = P^{-1}$) and $PA = LU$. A **permutation matrix** P has the rows of I in any order (there are $n!$ such matrices of order n). Because the rows of P are orthonormal unit vectors, $PP^T = I$, which gives $\boxed{P^{-1} = P^T}$. If A is any invertible matrix, a permutation P reorders its rows in advance so that no zero appears in a pivot position [Book: Eq. (8), p. 112]:

$$\boxed{PA = LU} \qquad (\text{or } A = L_1 P_1 U_1 \text{ if row exchanges are held until after elimination}). \tag{2.38}$$

2.7.2 Verbatim Book Examples and Worked Examples (Section 2.7)

BOOK EXAMPLE 2.7 A Recovering Symmetry with $PAQ = PAP^T$

Applying the permutation $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ to the rows of $A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 6 \\ 5 & 6 & 3 \end{bmatrix}$ gives $PA = \begin{bmatrix} 4 & 2 & 6 \\ 5 & 6 & 3 \\ 1 & 4 & 5 \end{bmatrix}$, which destroys its symmetry. What permutation Q applied to the *columns* of PA will recover symmetry in PAQ? Show that Q is P^T.

BOOK EXAMPLE 2.7 B Symmetric Factorization $A = LDL^T$

Find the symmetric factorization $A = LDL^T$ for the matrix $A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 6 \\ 5 & 6 & 3 \end{bmatrix}$ above. Is this A invertible? Find also the $PQ = LU$ factorization for $Q = P^T$.

BOOK EXAMPLE 2.7 C Block Saddle-Point Matrix $S = LDL^T$

For a rectangular A, this **saddle-point matrix** $S = \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} = S^T$ has size $m + n$. Apply block elimination to find a **block factorization** $S = LDL^T$. Then test invertibility:

$$S \text{ is invertible} \iff A^TA \text{ is invertible} \iff A\mathbf{x} \neq \mathbf{0} \text{ whenever } \mathbf{x} \neq \mathbf{0}.$$

SOLUTION AT END OF PAGE (Worked Examples 2.7 A, 2.7 B, and 2.7 C)

2.7 A: To put the original diagonal entries $2, 3, 1$ back onto the main diagonal, move column 2 to col 1, col 3 to col 2, and col 1 to col 3 using $Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = P^T = P^{-1}$. Then $PAQ = PAP^T = \begin{bmatrix} 2 & 6 & 4 \\ 6 & 3 & 5 \\ 4 & 5 & 1 \end{bmatrix}$ is symmetric.

2.7 B: Elimination on A uses multipliers $\ell_{21} = 4, \ell_{31} = 5, \ell_{32} = 1$ and yields pivots $1, -14, -8$:

$$A = LDL^T = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 5 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -14 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

A is invertible ($\det A = 1(-14)(-8) = 112 \neq 0$). For Q, $P = Q^{-1} = Q^T$ gives $PQ = I = (I)(I) = LU$.

2.7 C: Subtract A^T times block row 1 from block row 2 to get block $U = \begin{bmatrix} I & A \\ 0 & -A^TA \end{bmatrix}$.

Factoring out $D = \begin{bmatrix} I & 0 \\ 0 & -A^TA \end{bmatrix}$ gives:

$$S = LDL^T = \begin{bmatrix} I & 0 \\ A^T & I \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & -A^TA \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix}.$$

Since L is invertible, S is invertible iff D is invertible, which happens iff A^TA is invertible (independent columns of A).

2.7.3 Problem Set 2.7: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 2.7 (1–21) Transposes, Permutations, and Symmetric Matrices

1. Find $A^T, A^{-1}, (A^{-1})^T, (A^T)^{-1}$ for $A = \begin{bmatrix} 1 & 0 \\ 9 & 3 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & c \\ c & 0 \end{bmatrix}$.

2. Verify $(AB)^T = B^T A^T \neq A^T B^T$ for $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. When does $B^T A^T = A^T B^T$?

3. (a) $((AB)^{-1})^T$ comes from $(A^{-1})^T$ and $(B^{-1})^T$ in what order? (b) If U is upper triangular then $(U^{-1})^T$ is ____ triangular.

4. Show $A^2 = 0$ is possible but $A^T A = 0$ is impossible unless $A = 0$.

5. Compute $\mathbf{x}^T A\mathbf{y} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{x}^T A$, and $A\mathbf{y}$.

6. Transpose the block matrix $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$. When is M symmetric?

7. True or false: (a) $\begin{bmatrix} 0 & A \\ A & 0 \end{bmatrix}$ is automatically symmetric. (b) If A, B are symmetric then AB is symmetric. (c) If A is not symmetric then A^{-1} is not symmetric. (d) When A, B, C are symmetric, $(ABC)^T = CBA$.

8–10. **8:** Why are there $n!$ permutation matrices of order n? **9:** Give examples with $P_1 P_2 \neq P_2 P_1$ and $P_3 P_4 = P_4 P_3$. **10:** List all 12 "even" permutations of $(1, 2, 3, 4)$.

11. Which permutation makes PA upper triangular for $A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$? Which P_1, P_2 make $P_1 A P_2$ lower triangular?

12–15. **12:** Explain $(P\mathbf{x})^T (P\mathbf{y}) = \mathbf{x}^T \mathbf{y} \implies P^T P = I$. **13:** Find 3×3 $P \neq I$ with $P^3 = I$, and 4×4 $\widehat{P}$ with $\widehat{P}^4 \neq I$. **14:** Describe PAP for antidiagonal P. **15:** If P sends row 1 to row 4, P^T sends row ___ to row ___.

16. If $A = A^T$ and $B = B^T$, which are certainly symmetric: (a) $A^2 - B^2$, (b) $(A + B)(A - B)$, (c) ABA, (d) $ABAB$?

17. Find 2×2 symmetric matrices: (a) not invertible, (b) invertible but no LU without row exchange, (c) LDL^T has negative D.

18–21. **18:** Independent entries in 5×5 symmetric vs. skew-symmetric ($A^T = -A$). **19:** Symmetry of $R^T A R$ and positive diagonal of $R^T R$. **20:** Factor $\begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}, \begin{bmatrix} 1 & b \\ b & c \end{bmatrix}, K_3$ into LDL^T.

21: Find the 2×2 symmetric Schur complement of $\begin{bmatrix} 2 & 4 & 8 \\ 4 & 3 & 9 \\ 8 & 9 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$.

SOLUTION AT END OF PAGE (Problem Set 2.7 (Problems 1–21))

1. First A: $A^T = \begin{bmatrix} 1 & 9 \\ 0 & 3 \end{bmatrix}, A^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1/3 \end{bmatrix}, (A^{-1})^T = (A^T)^{-1} = \begin{bmatrix} 1 & -3 \\ 0 & 1/3 \end{bmatrix}$. Second A:

$A^T = A, A^{-1} = (A^{-1})^T = \frac{1}{c^2}\begin{bmatrix} 0 & c \\ c & -1 \end{bmatrix}$.

2. $(AB)^T = B^T A^T = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \neq A^T B^T = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. If $AB = BA$, then $B^T A^T = (AB)^T = (BA)^T = A^T B^T$.

3. (a) $(A^{-1})^T (B^{-1})^T$. (b) **Lower** triangular. **4.** $(A^T A)_{ii} = \|\text{col } i \text{ of } A\|^2 = 0 \implies A = 0$.

5. (a) 5 (the $(2,2)$ entry of A), (b) $\mathbf{x}^T A = [4 \;\; 5 \;\; 6]$, (c) $A\mathbf{y} = (2,5)^T$.

6. $M^T = \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix}$. Symmetric when $A = A^T, D = D^T$, and $B = C^T$.

7. (a) **False** (needs $A = A^T$), (b) **False**, (c) **True**, (d) **True**.

8–11. **8:** n choices for row 1, $n-1$ for row 2, ..., 1 for row n. **10:** $(1,2,3,4), (4,3,2,1), (2,1,4,3), (3,4,1,2)$ and the 8 cyclic 3-cycles $(2,3,1,4), (3,1,2,4), \ldots$ **11:** $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$; for lower triangular, $P_1 = I$ and $P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (exchanges **columns**).

12–17. **13:** (a) $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$; (b) $\widehat{P} = \text{diag}(P, 1)$ has $\widehat{P}^3 = I \implies \widehat{P}^4 = \widehat{P} \neq I$. **14:** PAP reverses the order of both rows and columns. **15:** Row **4** to row **1**. **16:** (a) and (c). **17:** (a) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, (b) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, (c) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

18–21. **18:** (a) $5+4+3+2+1 = 15$, (b) 10 in $L + 5$ in $D = 15$, (c) 10 (diagonal must be 0). **20:** $\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -7 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & c-b^2 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$. **21:** $\begin{bmatrix} -5 & -7 \\ -7 & -32 \end{bmatrix}$ and $\begin{bmatrix} d-b^2 & e-bc \\ e-bc & f-c^2 \end{bmatrix}$.

BOOK EXAMPLE PS 2.7 (22–40) $PA = LU$, **Networks, Matrix Groups, and Challenge Problems**

22. Find the $PA = LU$ factorizations for $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

23. Find a 4×4 permutation matrix A that needs 3 row exchanges to reach the end of elimination, and find P, L, U.

24. Factor $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 2 & 1 & 1 \end{bmatrix}$ into $PA = LU$ and $A = L_1 P_1 U_1$.

27. For $A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 11 & 4 \\ 0 & 4 & 9 \end{bmatrix}$, find E_{21} and E_{32} so $E_{32}E_{21}AE_{21}^T E_{32}^T = D = \text{diag}(1,2,1)$, and find L.

30. For the 3-city network $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}$ and $\mathbf{y} = (y_{BC}, y_{CS}, y_{BS})^T$: (a) Find $A^T\mathbf{y}$. (b) Verify $(A\mathbf{x})^T\mathbf{y} = \mathbf{x}^T(A^T\mathbf{y})$.

31–32. x_1 trucks and x_2 planes need $x_1 + 50x_2$ tons of steel (\$700/ton), $40x_1 + 1000x_2$ lb of rubber (\$3/lb), and $2x_1 + 50x_2$ months of labor (\$3000/month). Find A, $A^T\mathbf{y}$, and interpret $(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A^T\mathbf{y})$.

33–40. 33: Angle of rotation from $\mathbf{v} = (2,3,-5)$ to $P\mathbf{v} = (-5,2,3)$. **34:** Factor $\begin{bmatrix} 1 & 2 \\ 4 & 9 \end{bmatrix} = EH$ (elementary times symmetric). **36:** Which sets are matrix groups? **38:** Find a 5×5 permutation P with smallest period $P^6 = I$. **39:** Split $A = B + C$ into symmetric $B = \frac{1}{2}(A + A^T)$ and anti-symmetric $C = \frac{1}{2}(A - A^T)$. **40:** Orthogonal matrices $Q^TQ = I$.

SOLUTION AT END OF PAGE (Problem Set 2.7 (Problems 22–40))

22. First A: $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$. Second A: $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

23. $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ has $P = A^T$, $L = I$, $U = I$.

24. $PA = LU$: $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$. $A = L_1P_1U_1$: $L_1 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, U_1 = U$.

27. $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}, L = (E_{32}E_{21})^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$.

30. (a) $A^T\mathbf{y} = (y_{BC} + y_{BS}, -y_{BC} + y_{CS}, -y_{CS} - y_{BS})^T$. (b) Both equal $(x_B - x_C)y_{BC} + (x_C - x_S)y_{CS} + (x_B - x_S)y_{BS}$.

31–32. $A = \begin{bmatrix} 1 & 50 \\ 40 & 1000 \\ 2 & 50 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 700 \\ 3 \\ 3000 \end{bmatrix} \implies A^T\mathbf{y} = \begin{bmatrix} 6820 \\ 188000 \end{bmatrix}$ (value of 1 truck is \$6,820; 1 plane is \$188,000). $(A\mathbf{x}) \cdot \mathbf{y}$ is the total cost of inputs; $\mathbf{x} \cdot (A^T\mathbf{y})$ is the total value of outputs.

33–40. **33:** $\cos\theta = \frac{\mathbf{v} \cdot P\mathbf{v}}{\|\mathbf{v}\| \, \|P\mathbf{v}\|} = \frac{-10+6-15}{38} = -\frac{1}{2} \implies \theta = 120^\circ$. **34:** $\begin{bmatrix} 1 & 2 \\ 4 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$.

36: Groups: Lower triangular L with 1s on diagonal, diagonal invertible D, permutations P, orthogonal $Q^T = Q^{-1}$. **38:** Block diagonal $P = \text{diag}(P_{2\times2}, P_{3\times3})$ has period $\text{lcm}(2,3) = 6$.

39: $B = \frac{1}{2}(A + A^T)$ and $C = \frac{1}{2}(A - A^T)$. **40:** $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

Chapter 2: Unique Synthesis Problems & Formula Sheet

Chapter 2 Core Equations (Derived from +2 Base)

- **Row vs. Column Multiplication:** $(A\mathbf{x})_i = \sum_{j=1}^{n} a_{ij}x_j = (\text{row } i)\cdot\mathbf{x}$ and $A\mathbf{x} = \sum_{j=1}^{n} x_j(\text{column } j)$.
- **Elimination Multiplier:** $\ell_{ij} = \dfrac{\text{entry to eliminate in row } i}{\text{pivot in row } j}$ (Pivots d_k can never be zero).
- **Four Ways to Multiply** AB**:** (1) $A[\mathbf{b}_1 \dots \mathbf{b}_p]$, (2) $[(\text{row } i \text{ of } A)B]$, (3) $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$, (4) $\sum_{k=1}^{n}(\text{col } k \text{ of } A)(\text{row } k \text{ of } B)$.
- **Block Schur Complement:** $\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & S \end{bmatrix}$ where $S = D - CA^{-1}B$.
- **Inverses & Transposes:** $(AB)^{-1} = B^{-1}A^{-1}$, $(AB)^T = B^TA^T$, $(A^{-1})^T = (A^T)^{-1}$, $(A\mathbf{x})^T\mathbf{y} = \mathbf{x}^T(A^T\mathbf{y})$.
- **Factorizations:** $PA = LU$, $A = LDU$, and for symmetric $A = A^T$: $A = LDL^T$.
- **Operation Counts:** Factor $A = LU$: $\frac{1}{3}(n^3 - n)$ ops ($\frac{1}{6}n^3$ if symmetric, nw^2 if banded). Solve $L\mathbf{c} = \mathbf{b}, U\mathbf{x} = \mathbf{c}$: n^2 ops ($2nw$ if banded).

BOOK EXAMPLE Ch2-U1 Unique Synthesis Problem: Tridiagonal Inverse & Woodbury Formula

(a) For the $n \times n$ second-difference matrix $K_n = \text{tridiag}(-1, 2, -1)$, derive the explicit (i, j) entry of K_n^{-1} and compare it with T_n^{-1} where $T_{11} = 1$ instead of 2.

(b) Use the rank-one Woodbury-Morrison formula $(K_n - \mathbf{e}_1\mathbf{e}_1^T)^{-1} = K_n^{-1} + \frac{K_n^{-1}\mathbf{e}_1\mathbf{e}_1^TK_n^{-1}}{1-\mathbf{e}_1^TK_n^{-1}\mathbf{e}_1}$ to get T_n^{-1} directly from K_n^{-1} for $n = 3$.

SOLUTION AT END OF PAGE (Chapter 2 Unique Synthesis Problem Ch2-U1)

(a) From Problem 2.6.26, $(n+1)K_n^{-1}$ has first row $(n, n-1, \dots, 1)$, and for $i \le j$, row i is i times row 1, so:

$$(K_n^{-1})_{ij} = \frac{\min(i,j)\big(n+1-\max(i,j)\big)}{n+1}, \qquad (T_n^{-1})_{ij} = n+1-\max(i,j).$$

(b) For $n = 3$, $T_3 = K_3 - \mathbf{e}_1\mathbf{e}_1^T$. The first column of K_3^{-1} is $\mathbf{c}_1 = \frac{1}{4}(3,2,1)^T$, and $\mathbf{e}_1^TK_3^{-1}\mathbf{e}_1 = \frac{3}{4}$. The Woodbury formula gives:

$$T_3^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} + \frac{1}{1-3/4}\left(\frac{1}{16}\begin{bmatrix} 9 & 6 & 3 \\ 6 & 4 & 2 \\ 3 & 2 & 1 \end{bmatrix}\right) = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

CHAPTER 3

Complete Solution to $Ax = b$ and Subspaces

Chapter 3 Syllabus: 3.4 The Complete Solution to $A\mathbf{x} = \mathbf{b}$ (pp. 155–167) · 3.6 Dimensions of the Four Subspaces (pp. 193–194).

3.1 The Complete Solution to $Ax = b$ [Book 3.4, pp. 155–167]

3.1.1 Derivation 8: Augmented Row Reduction $[A \;\; \mathbf{b}] \to [R \;\; \mathbf{d}]$ and $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$

GOAL

Derive the complete solution $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ to any linear system $A\mathbf{x} = \mathbf{b}$ using the row reduced echelon form $[R \;\; \mathbf{d}]$, and classify the four solvability cases by rank r.

FROM YOUR +2 BASE

Use linearity $A(\mathbf{x}_p + \mathbf{x}_n) = A\mathbf{x}_p + A\mathbf{x}_n = \mathbf{b} + \mathbf{0} = \mathbf{b}$.

Step 1: Reduce $[A \;\; \mathbf{b}]$ to $[R \;\; \mathbf{d}]$ and find the solvability condition. Suppose A is $m \times n$ with rank r (r pivots). Gauss-Jordan elimination on $[A \;\; \mathbf{b}]$ produces $[R \;\; \mathbf{d}]$, where R has r nonzero rows (with I_r in the r pivot columns) and $m - r$ zero rows at the bottom.

$$\text{For } A\mathbf{x} = \mathbf{b} \text{ to be solvable, the last } m - r \text{ components of } \mathbf{d} \text{ must equal } 0 \quad (0 = 0). \tag{3.1}$$

Step 2: Construct the particular solution $\mathbf{x}_p$ and nullspace solution $\mathbf{x}_n$. Set all $n - r$ **free variables** equal to zero. Because the r pivot columns of R form the identity matrix I_r, the r **pivot variables** in $\mathbf{x}_p$ equal the first r entries of $\mathbf{d}$:

$$\boxed{A\mathbf{x}_p = \mathbf{b} \quad (\text{free variables} = 0, \;\; \text{pivot variables from } \mathbf{d})}. \tag{3.2}$$

For the homogeneous equation $A\mathbf{x}_n = \mathbf{0}$ (or $R\mathbf{x}_n = \mathbf{0}$), there are $n - r$ **special solutions** $\mathbf{s}_1, \ldots, \mathbf{s}_{n-r}$, obtained by setting one free variable to 1 and the other free variables to 0.

Step 3: Add $\mathbf{x}_p$ and $\mathbf{x}_n$ for the complete solution. Every solution to $A\mathbf{x} = \mathbf{b}$ is the sum of one particular solution $\mathbf{x}_p$ and any nullspace vector $\mathbf{x}_n$:

$$\boxed{\mathbf{x}_{\text{complete}} = \mathbf{x}_p + \mathbf{x}_n = \mathbf{x}_p + c_1\mathbf{s}_1 + c_2\mathbf{s}_2 + \cdots + c_{n-r}\mathbf{s}_{n-r}}. \tag{3.3}$$

Complete Solution x = x_p + x_n (Parallel Affine Line)

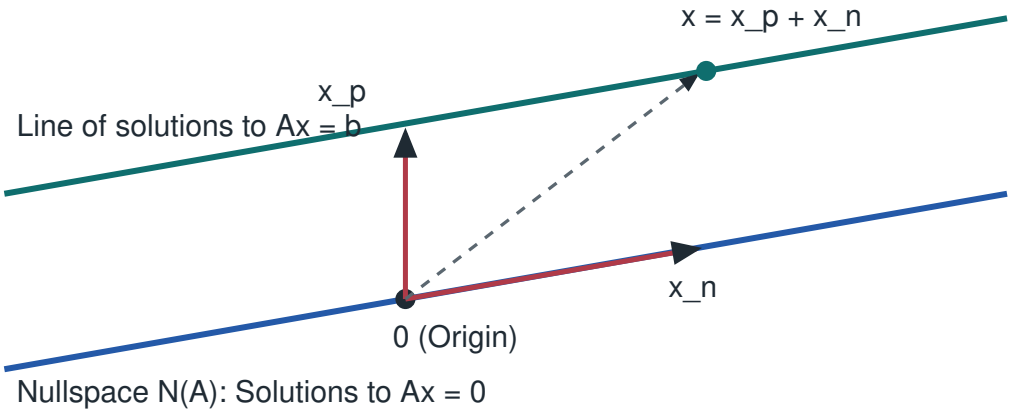


Fig. 3.1 Geometric meaning of $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$: the nullspace $N(A)$ goes through the origin $\mathbf{0}$, and $\mathbf{x}_p$ shifts that subspace to form the parallel affine solution set for $A\mathbf{x} = \mathbf{b}$.

RESULTThe Four Possibilities for Linear Equations [Book p. 159]

Case	Rank r	Shape	Reduced R	Number of Solutions to $A\mathbf{x} = \mathbf{b}$
1. Full rank	$r = m = n$	Square invertible	$R = [I]$	**Exactly** 1 solution for every $\mathbf{b}$
2. Full row rank	$r = m < n$	Short and wide	$R = [I \;\; F]$	∞ **solutions** for every $\mathbf{b}$
3. Full col rank	$r = n < m$	Tall and thin	$R = \begin{bmatrix} I \\ 0 \end{bmatrix}$	0 **or** 1 solution
4. Not full rank	$r < m, r < n$	General	$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$	0 **or** ∞ solutions

3.1.2 Verbatim Book Examples and Worked Examples (Section 3.4)

BOOK EXAMPLE 3.4-Ex1–2 Full Column Rank and Full Row Rank

Example 1 (Full column rank): Find the condition on (b_1, b_2, b_3) for $A\mathbf{x} = \mathbf{b}$ to be solvable, and find $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$, if $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ -2 & -3 \end{bmatrix}$.

Example 2 (Full row rank): Find $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ for $x + y + z = 3, x + 2y - z = 4$.

BOOK EXAMPLE 3.4 A Complete Six-Step Analysis of $A\mathbf{x} = \mathbf{b}$

This question connects elimination to column space-nullspace-rank-solvability. A has rank 2:

$$A\mathbf{x} = \mathbf{b} \quad \text{is} \quad \begin{aligned} x_1 + 2x_2 + 3x_3 + 5x_4 &= b_1 \\ 2x_1 + 4x_2 + 8x_3 + 12x_4 &= b_2 \\ 3x_1 + 6x_2 + 7x_3 + 13x_4 &= b_3. \end{aligned}$$

1. Reduce $[A\ \ \mathbf{b}]$ to $[U\ \ \mathbf{c}]$. 2. Find the condition on b_1, b_2, b_3 for $A\mathbf{x} = \mathbf{b}$ to have a solution. 3. Describe the column space of A. 4. Describe the nullspace of A. 5. Find $\mathbf{x}_p$ and the complete solution for $\mathbf{b} = (0, 6, -6)$. 6. Reduce $[U\ \ \mathbf{c}]$ to $[R\ \ \mathbf{d}]$.

BOOK EXAMPLE 3.4 B–C Deductions from Solutions and Left Nullspace Condition

3.4 B: What does each condition tell you about A and $\mathbf{b}$? (1) Exactly one solution. (2) All solutions have the form $\mathbf{x} = \begin{bmatrix}2\\1\end{bmatrix} + c\begin{bmatrix}1\\1\end{bmatrix}$. (3) No solutions. (4) $\mathbf{x} = \begin{bmatrix}1\\1\\0\end{bmatrix} + c\begin{bmatrix}1\\0\\1\end{bmatrix}$. (5) Infinitely many solutions.

3.4 C: Find $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ for $\begin{bmatrix}1&2&1&0\\2&4&4&8\\4&8&6&8\end{bmatrix}\mathbf{x} = \begin{bmatrix}4\\2\\10\end{bmatrix}$ and find y_1, y_2, y_3 so $y_1(\text{row } 1) + y_2(\text{row } 2) + y_3(\text{row } 3) = \mathbf{0}$.

SOLUTION AT END OF PAGE (Examples 3.4-Ex1–2, 3.4 A, 3.4 B, and 3.4 C)

3.4-Ex1–2: In Ex. 1, $[A\ \ \mathbf{b}] \to \begin{bmatrix}1&0&2b_1-b_2\\0&1&b_2-b_1\\0&0&b_1+b_2+b_3\end{bmatrix}$, so $b_1+b_2+b_3=0$ and $\mathbf{x} = \begin{bmatrix}2b_1-b_2\\b_2-b_1\end{bmatrix} + \begin{bmatrix}0\\0\end{bmatrix}$. In Ex. 2, $[A\ \ \mathbf{b}] \to [R\ \ \mathbf{d}] = \begin{bmatrix}1&0&3&2\\0&1&-2&1\end{bmatrix} \Longrightarrow \mathbf{x} = \begin{bmatrix}2\\1\\0\end{bmatrix} + x_3\begin{bmatrix}-3\\2\\1\end{bmatrix}$.

3.4 A: (1) $[U\ \ \mathbf{c}] = \left[\begin{array}{cccc|c}1&2&3&5&b_1\\0&0&2&2&b_2-2b_1\\0&0&0&0&b_3+b_2-5b_1\end{array}\right]$. (2) Solvable iff $b_3+b_2-5b_1=0$. (3) $C(A)$ is the plane through $(1,2,3)$ and $(3,8,7)$ described by $b_3+b_2-5b_1=0$. (4) Special solutions: $\mathbf{s}_1 = (-2,1,0,0)^T$ and $\mathbf{s}_2 = (-2,0,-1,1)^T$. (5) $\mathbf{x}_p = (-9,0,3,0)^T$, $\mathbf{x} = \mathbf{x}_p + c_1\mathbf{s}_1 + c_2\mathbf{s}_2$. (6) $[R\ \ \mathbf{d}] = \left[\begin{array}{cccc|c}1&2&0&2&-9\\0&0&1&1&3\\0&0&0&0&0\end{array}\right]$.

3.4 B–C: 3.4 B: (1) $r = n \le m$; (2) $n = 2, r = 1, \text{col } 2 = -\text{col } 1$; (3) $r < m, \mathbf{b} \neq \mathbf{0}$; (4) $n = 3, r = 2 \le m, \text{col } 3 = -\text{col } 1, \mathbf{b} = \text{col } 1 + \text{col } 2$; (5) $r < n$. **3.4 C:** $\mathbf{x}_p = (7,0,-3,0)^T$, $\mathbf{s}_1 = (-2,1,0,0)^T$, $\mathbf{s}_2 = (4,0,-4,1)^T$; $\mathbf{y} = (2,1,-1)$ satisfies $2(4)+1(2)-1(10)=0$.

3.1.3 Problem Set 3.4 & 3.6: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 3.4 (1–18) Complete Solutions, Solvability Conditions, and Full Rank

1. Execute the six steps of Worked Example 3.4 A for $A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$.

2. Carry out the six steps for rank-one $A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ 4 & 2 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 10 \\ 30 \\ 20 \end{bmatrix}$.

3–4. Find the complete solution to: (3) $x + 3y + 3z = 1, 2x + 6y + 9z = 5, -x - 3y + 3z = 5$; (4) $\begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.

5. Under what condition on b_1, b_2, b_3 is $x + 2y - 2z = b_1, 2x + 5y - 4z = b_2, 4x + 9y - 8z = b_3$ solvable? Find all solutions when that condition holds.

6. What conditions on b_1, b_2, b_3, b_4 make $\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 2 & 5 \\ 3 & 9 \end{bmatrix} \mathbf{x} = \mathbf{b}$ and $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 12 \end{bmatrix} \mathbf{x} = \mathbf{b}$ solvable? Find $\mathbf{x}$.

7–8. 7: Show $(b_1, b_2, b_3) \in C(A)$ for $A = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 8 & 2 \\ 2 & 4 & 0 \end{bmatrix}$ iff $b_3 - 2b_2 + 4b_1 = 0$. **8:** Find the column space conditions for (a) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 3 \\ 0 & 2 & 5 \end{bmatrix}$ and (b) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}$.

9–15. 10: Construct a 2×3 system with $\mathbf{x}_p = (2, 4, 0)$ and $\mathbf{x}_n = c(1, 1, 1)$. **11:** Why can't a 1×3 system have $\mathbf{x}_n = c(1, 1, 1)$? **12:** If $A\mathbf{x} = \mathbf{b}$ has $\mathbf{x}_1, \mathbf{x}_2$, find solutions to $A\mathbf{x} = \mathbf{0}$ and $A\mathbf{x} = \mathbf{b}$. **13:** Explain why 13(a)–(d) are false. **14–15:** Fill in blanks on pivotless column 5 and pivotless row 3 of U.

16–18. 16: Largest rank of 3×5 matrix is ___. **17:** Largest rank of 6×4 matrix is ___. **18:** Find rank of $\begin{bmatrix} 1 & 4 & 0 \\ 2 & 11 & 5 \\ -1 & 2 & 10 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & q \end{bmatrix}$.

SOLUTION AT END OF PAGE (Problem Set 3.4 (Problems 1–18))

1. $[U \;\; \mathbf{c}] = \left[\begin{array}{cccc|c} 2 & 4 & 6 & 4 & b_1 \\ 0 & 1 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 2b_1 \end{array}\right]$. Condition: $b_3 + b_2 - 2b_1 = 0$. $\mathbf{x}_p = (4, -1, 0, 0)^T$, $\mathbf{s}_1 = (-1, -1, 1, 0)^T$, $\mathbf{s}_2 = (2, -2, 0, 1)^T$.

2. Conditions: $b_2 = 3b_1$ and $b_3 = 2b_1$ (puts $\mathbf{b}$ into the **column** space, a line). $\mathbf{x}_p = (5, 0, 0)^T$, $\mathbf{s}_1 = (-1/2, 1, 0)^T$, $\mathbf{s}_2 = (-3/2, 0, 1)^T$.

3–4. (3) $\mathbf{x} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$. (4) $\mathbf{x} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \\ 0 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$.
5. Solvable iff $b_3 - b_2 - 2b_1 = 0$. Then $\mathbf{x} = \begin{bmatrix} 5b_1 - 2b_2 \\ b_2 - 2b_1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$.
6. Both require $b_2 = 2b_1$ and $b_4 - 3b_3 + 3b_1 = 0$. First: $\mathbf{x} = (5b_1 - 2b_3, b_3 - 2b_1)^T$. Second: add $x_3(-1, -1, 1)^T$.
7–8. 7: $4(\text{row } 1) - 2(\text{row } 2) + (\text{row } 3) = \mathbf{0}$. **8:** (a) Invertible ($r = 3$), all $\mathbf{b} \in \mathbb{R}^3$ in $C(A)$; (b) $b_3 = 2b_2$.
10–18. 10: $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. **11:** A 1×3 nonzero matrix has $3 - 1 = 2$ free variables (a plane of nullspace solutions, not a line). **12: 0** and $\mathbf{x}_1 - \mathbf{x}_2$ solve $A\mathbf{x} = \mathbf{0}$; $2(\mathbf{x}_1 - \mathbf{x}_2)$ solves $A\mathbf{x} = \mathbf{0}$ and $\frac{1}{2}(\mathbf{x}_1 + \mathbf{x}_2)$ solves $A\mathbf{x} = \mathbf{b}$. **14: free**; **is not**; **infinitely many**. **15: all zeros**; $c_3 = 0$; **might not be**. **16: 3**; **row**; **always exists**; $\mathbb{R}^3$. **17: 4**; **column**; **is unique**; $\{\mathbf{0}\}$. **18:** First matrix has rank 2; second has rank 2 if $q = 2$ and rank 3 if $q \neq 2$.

BOOK EXAMPLE PS 3.4 (19–36) & PS 3.6 (19–32) Reduced Echelon R and the Four Subspaces

3.4.19–25. 19: Rank of A, A^TA, AA^T. **21:** Complete solution to (a) $x+y+z=4$, (b) $x+y+z=4, x-y+z=4$. **23:** Choose q for ranks $1,2,3$ on $A=\begin{bmatrix}6&4&2\\-3&-2&-1\\9&6&q\end{bmatrix}$ and $B=\begin{bmatrix}3&1&3\\q&2&q\end{bmatrix}$. **25:** Relations between r,m,n for (a) no solution for some $\mathbf{b}$, (b) ∞ solutions for every $\mathbf{b}$, (c) 1 or 0 solutions, (d) 1 solution for every $\mathbf{b}$.

3.4.26–36. 30: Reduce $\begin{bmatrix}1&0&2&3\\1&3&2&0\\2&0&4&9\end{bmatrix}\mathbf{x}=\begin{bmatrix}2\\5\\10\end{bmatrix}$ to $U\mathbf{x}=\mathbf{c}$ and $R\mathbf{x}=\mathbf{d}$, and find $\mathbf{x}_p, \mathbf{x}_n$. **32:** Find LU and complete solutions for $A=\begin{bmatrix}1&3&1\\1&2&3\\2&4&6\\1&1&5\end{bmatrix}$ with $\mathbf{b}=(1,3,6,5)^T$ and $\mathbf{b}=(1,0,0,0)^T$. **34:** 3×4 matrix A has $\mathbf{s}=(2,3,1,0)^T$ as its only special solution: find rank, R, and why $A\mathbf{x}=\mathbf{b}$ is always solvable.

3.6.19–25. 19: Find basis vectors in the left nullspace $N(A^T)$ for (a) $\begin{bmatrix}1&2\\3&4\\4&6\end{bmatrix}$, (b) $\begin{bmatrix}1&2\\2&3\\2&4\\2&5\end{bmatrix}$. **20:** For $A=ER=\begin{bmatrix}1&0&0\\2&1&0\\3&4&1\end{bmatrix}\begin{bmatrix}4&2&0&1\\0&0&1&3\\0&0&0&0\end{bmatrix}$, why is $\mathbf{y}^T$ in $N(A^T)$ the last row of E^{-1}? **21–22:** Rank-two matrix $A=\mathbf{u}\mathbf{v}^T+\mathbf{w}\mathbf{z}^T$. **24:** $A^T\mathbf{y}=\mathbf{d}$ is solvable when $\mathbf{d}$ is in which subspace, and unique when?

3.6.26–32. 26: Prove $\text{rank}(AB)\le\text{rank}(A)$ and $\text{rank}(AB)\le\text{rank}(B)$. **28:** Find ranks and bases for the 8×8 checkerboard matrix B and chess matrix C. **31:** For $A=\begin{bmatrix}1&0&-1\\-1&1&0\\0&-1&1\end{bmatrix}$ multiplying 3×3 matrices X, find dimensions of $\{X: AX=0\}$ and $\{AX\}$. **32:** If $A=\begin{bmatrix}I&F\\0&0\end{bmatrix}$ and $B=\begin{bmatrix}I&G\\0&0\end{bmatrix}$ have the same four subspaces, prove $F=G$.

SOLUTION AT END OF PAGE (Problem Set 3.4 (19–36) and Problem Set 3.6 (19–32))

3.4.19–25. 19: $\text{rank}(A)=\text{rank}(A^TA)=\text{rank}(AA^T)=2$ for both. **21:** (a) $\mathbf{x}=(4,0,0)^T+y(-1,1,0)^T+z(-1,0,1)^T$; (b) $\mathbf{x}=(4,0,0)^T+z(-1,0,1)^T$. **23:** For A: rank 1 if $q=3$, rank 2 if $q\neq 3$, rank 3 impossible. For B: rank 1 if $q=6$, rank 2 if $q\neq 6$, rank 3 impossible. **25:** (a) $r<m$; (b) $r=m<n$; (c) $r=n<m$; (d) $r=m=n$.

3.4.30–34. 30: $[R\ \ \mathbf{d}]=\left[\begin{array}{cccc|c}1&0&2&0&-4\\0&1&0&0&3\\0&0&0&1&2\end{array}\right]\implies \mathbf{x}_p=(-4,3,0,2)^T, \mathbf{x}_n=x_3(-2,0,1,0)^T$. **32:** With $\mathbf{b}=(1,3,6,5)^T$, $\mathbf{x}=(7,-2,0)^T+x_3(-7,2,1)^T$; with $\mathbf{b}=(1,0,0,0)^T$, no solution. **34:**

(a) $r = 4 - 1 = 3$, $\mathbf{x}_n = c(2,3,1,0)^T$; (b) $R = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$; (c) $r = m = 3$ (full row rank).

3.6.19–25. 19: (a) $\mathbf{y} = (-1,-1,1)^T$; (b) $\mathbf{y}_1 = (0,-2,1,0)^T, \mathbf{y}_2 = (1,-3,0,1)^T$. **20:** $E^{-1}A = R$ has zero 3rd row, so (row 3 of E^{-1})$A = [1 \quad -4 \quad 1]A = \mathbf{0}^T$. **24:** Solvable when $\mathbf{d} \in C(A^T)$ (the **row space**); unique when $N(A^T)$ (the **left nullspace**) contains only **0**.

3.6.26–32. 26: Fill in: B, B, A. **28:** Checkerboard B has rank 2; chess matrix C has rank 2. **31:** $\text{rank}(A) = 2$, so each of the 3 columns of X has 1 free variable: $\dim\{X : AX = 0\} = 3(1) = 3$ and $\dim\{AX\} = 3(2) = 6$ ($3 + 6 = 9$). **32:** Same nullspace means the special solutions $\begin{bmatrix} -F \\ I \end{bmatrix}$ and $\begin{bmatrix} -G \\ I \end{bmatrix}$ are identical, so $F = G$.

Chapter 3: Unique Synthesis Problems & Formula Sheet

Chapter 3 Core Equations (Derived from +2 Base)

- **Complete Solution to $A\mathbf{x} = \mathbf{b}$:** $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n = \mathbf{x}_p + c_1\mathbf{s}_1 + \cdots + c_{n-r}\mathbf{s}_{n-r}$.
- **Row Reduced Form $R = \mathbf{rref}(A)$:** If $R = \begin{bmatrix} I_r & F \\ 0 & 0 \end{bmatrix}$, then the special solutions in $N(A)$ are columns of $N = \begin{bmatrix} -F \\ I_{n-r} \end{bmatrix}$, satisfying $RN = 0$.
- **Fundamental Theorem of Linear Algebra (Dimensions of the Four Subspaces):**

$$\dim C(A) = \dim C(A^T) = r, \qquad \dim N(A) = n - r, \qquad \dim N(A^T) = m - r.$$

- **Solvability & Left Nullspace Test:** $\mathbf{b} \in C(A) \iff \mathbf{y}^T\mathbf{b} = 0$ for every $\mathbf{y} \in N(A^T)$ (where $\mathbf{y}^T A = \mathbf{0}^T$).

BOOK EXAMPLE Ch3-U1 Unique Synthesis Problem: Parabolic Discrete Boundary Value Problem

Suppose K is the 9 by 9 second difference matrix (2's on the diagonal, -1's on the diagonal above and below). Solve the equation $K\mathbf{x} = \mathbf{b} = (10, \ldots, 10)^T$ and derive the exact quadratic parabola $x_k = p(k)$ that passes through all nine components $x_1, \ldots, x_9$ (with boundary values $x_0 = x_{10} = 0$).

SOLUTION AT END OF PAGE (Chapter 3 Unique Synthesis Problem Ch3-U1)

The kth equation of $K\mathbf{x} = (10, \ldots, 10)^T$ is the second-difference equation:

$$-x_{k-1} + 2x_k - x_{k+1} = 10 \quad \text{for } k = 1, 2, \ldots, 9, \quad \text{with } x_0 = 0 \text{ and } x_{10} = 0.$$

From +2 algebra, the second difference of k^2 is $-(k-1)^2 + 2k^2 - (k+1)^2 = -2$. To get $+10$ on the right side and vanish at $k = 0$ and $k = 10$, choose the parabola:

$$\boxed{x_k = 5k(10 - k)} \qquad (k = 1, 2, \ldots, 9).$$

Check: $-5(k-1)(11-k) + 10k(10-k) - 5(k+1)(9-k) = 5\big[-(k^2 - 12k + 11) + 2(10k - k^2) - (9 + 8k - k^2)\big] = 5(2) = 10$. Thus $\mathbf{x} = (45, 80, 105, 120, 125, 120, 105, 80, 45)^T$.

CHAPTER 4

Determinants

Chapter 5 Syllabus: 5.1 The Properties of Determinants (pp. 244–254) · 5.2 Permutations and Cofactors (pp. 255–268) · 5.3 Cramer's Rule, Inverses, and Volumes (pp. 269–282).

4.1 The Properties of Determinants [Book 5.1, pp. 244–254]

4.1.1 Derivation 9: Deriving Rules 4–10 from the Three Defining Axioms (Rules 1–3)

GOAL

Define the determinant of an $n \times n$ matrix by three simple axioms (Rules 1–3) and derive Rules 4–10 step-by-step.

FROM YOUR +2 BASE

Use **T6** ($\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$) to check every rule on 2×2 matrices.

THE IDEA

The Three Defining Axioms (Rules 1–3) [Book pp. 246]**:**

1. **Rule 1 (Identity):** $\det I = 1$.
2. **Rule 2 (Sign Reversal):** The determinant changes sign when two rows are exchanged ($\det P = \pm 1$).
3. **Rule 3 (Linearity in Each Row Separately):** Scaling one row by t multiplies $\det A$ by t: $\det[t \text{ row } 1; \dots] = t \det A$ (so $\boxed{\det(tA) = t^n \det A}$). Adding row $1'$ to row 1 (keeping all other rows fixed) adds the determinants.

Step 1: Derive Rules 4, 5, and 6 (Equal Rows, Elimination, and Zero Row).

- **Rule 4 (Equal Rows $\implies$ $\det A = 0$):** Exchange the two equal rows. By Rule 2, $\det A$ changes to $-\det A$, but the matrix is unchanged, so $\det A = -\det A \implies \boxed{\det A = 0}$.
- **Rule 5 (Elimination Leaves $\det A$ Unchanged):** Subtracting ℓ(row i) from row k splits by Rule 3 into $\det A - \ell \det(\text{matrix with row } k = \text{row } i) = \det A - \ell(0) = \boxed{\det A}$.
- **Rule 6 (Row of Zeros $\implies$ $\det A = 0$):** Multiply the zero row by $t = 0$ using Rule 3 (or add another row to it by Rule 5 to get two equal rows) to get $\boxed{\det A = 0}$.

Step 2: Derive Rules 7 and 8 (Triangular Matrices and Invertibility Test).

- **Rule 7 (Triangular Matrix):** Use Rule 5 to eliminate all off-diagonal entries, leaving $\text{diag}(a_{11}, \dots, a_{nn})$. Factor out $a_{11}, \dots, a_{nn}$ one row at a time by Rule 3 and apply Rule 1 ($\det I = 1$) to get:

$$\boxed{\det(\text{Triangular } A) = a_{11}a_{22} \cdots a_{nn} = \text{product of diagonal entries}}. \tag{4.1}$$

- **Rule 8 (Singular vs. Invertible):** Elimination takes $PA = LU$. If A is singular, U has a zero row so $\det A = 0$. If A is invertible, U has pivots $d_1, \ldots, d_n$ on its diagonal [Book: Eq. (2), p. 248]:

$$\boxed{\det A = \pm \det U = \pm(d_1 d_2 \cdots d_n) \neq 0}. \tag{4.2}$$

Step 3: Derive Rules 9 and 10 (Product Rule and Transpose Rule).

- **Rule 9 (Product Rule $\det(AB) = (\det A)(\det B)$):** For fixed invertible B, define $D(A) = \det(AB)/\det B$. (1) $D(I) = \det B/\det B = 1$ (Rule 1). (2) Exchanging two rows of A exchanges the same two rows of AB, reversing the sign of $D(A)$ (Rule 2). (3) Linear combinations in row 1 of A give the same linear combinations in row 1 of AB, so $D(A)$ is linear in row 1 (Rule 3). Thus $D(A) = \det A$, proving:

$$\boxed{\det(AB) = (\det A)(\det B)} \quad \text{and for } B = A^{-1}: \quad \boxed{\det(A^{-1}) = \frac{1}{\det A}}. \tag{4.3}$$

- **Rule 10 (Transpose Rule $\det A^T = \det A$):** Transpose $PA = LU$ to get $A^T P^T = U^T L^T$. Since $\det L = \det L^T = 1$, $\det U = \det U^T$, and $\det P = \det P^T = \pm 1$, Rule 9 gives $\boxed{\det A^T = \det A}$.

4.1.2 Verbatim Worked Examples (Section 5.1)

BOOK EXAMPLE 5.1 A Checkerboard Signs and Row Combinations

Apply these operations to A and find the determinants of M_1, M_2, M_3: In M_1, multiplying each a_{ij} by $(-1)^{i+j}$ gives a checkerboard sign pattern. In M_2, rows $1, 2, 3$ of A are *subtracted* from rows $2, 3, 1$. In M_3, rows $1, 2, 3$ of A are *added* to rows $2, 3, 1$. *How are the determinants of M_1, M_2, M_3 related to the determinant of A?*

BOOK EXAMPLE 5.1 B Eigenvalue Determinant by Row and Column Operations

Explain how to reach this determinant by row operations:

$$\det \begin{bmatrix} 1-a & 1 & 1 \\ 1 & 1-a & 1 \\ 1 & 1 & 1-a \end{bmatrix} = a^2(3-a).$$

SOLUTION AT END OF PAGE (Worked Examples 5.1 A and 5.1 B)

5.1 A: (1) $M_1 = DAD$ where $D = \operatorname{diag}(1, -1, 1)$ has $\det D = -1$. By Rule 9, $\det M_1 = (-1)(\det A)(-1) = \boxed{\det A}$. (2) In M_2, the sum of the three rows is the zero row, so M_2 is singular and $\boxed{\det M_2 = 0}$. (3) Splitting M_3 by linearity in each row (Rule 3) gives $2^3 = 8$ determinants; 6 have repeated rows (0), leaving $\det[\text{row } 1; \text{row } 2; \text{row } 3] + \det[\text{row } 3; \text{row } 1; \text{row } 2] = \det A + \det A = \boxed{2 \det A}$.

5.1 B: Subtract row 3 from row 1 and from row 2 (Rule 5) to get $\det \begin{bmatrix} -a & 0 & a \\ 0 & -a & a \\ 1 & 1 & 1-a \end{bmatrix}$.
Next, add column 1 and column 2 to column 3 (Rule 5 + Rule 10) to get a lower triangular matrix with diagonal entries $-a, -a, 3-a$. By Rule 7, the determinant is $(-a)(-a)(3-a) = \boxed{a^2(3-a)}$.

4.1.3 Problem Set 5.1: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 5.1 (1–34) Rules for Determinants and Row Operations

1. If a 4×4 matrix has $\det A = \frac{1}{2}$, find $\det(2A)$, $\det(-A)$, $\det(A^2)$, and $\det(A^{-1})$.
2. If a 3×3 matrix has $\det A = -1$, find $\det(\frac{1}{2}A)$, $\det(-A)$, $\det(A^2)$, and $\det(A^{-1})$.
3. True or false: (a) $\det(I + A) = 1 + \det A$. (b) $\det(ABC) = |A|\,|B|\,|C|$. (c) $\det(4A) = 4|A|$. (d) $\det(AB - BA) = 0$.
4–5. Count the row exchanges to permute the reverse identity J_n to I_n for $n = 3, 4, 5, 6, 7$, propose a general rule, and find $\det J_{101}$.

6–8. 6: Show Rule 6 comes from Rule 3. **7:** Find $\det Q$ for rotation $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and reflection $Q = \begin{bmatrix} 1 - 2\cos^2\theta & -2\cos\theta\sin\theta \\ -2\cos\theta\sin\theta & 1 - 2\sin^2\theta \end{bmatrix}$. **8:** Prove orthogonal $Q^TQ = I \implies \det Q = \pm 1$.

9–12. 9: Find $\det A, \det B, \det C$ for $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. **11:** What is the flaw in $CD = -DC \implies |C|\,|D| = -|D|\,|C|$? **12:** What is wrong with $\det\left(\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) = \frac{ad-bc}{ad-bc} = 1$?

13–19. Compute determinants by row operations: (13) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix}$; (14) $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 6 & 6 & 1 \\ -1 & 0 & 0 & 3 \\ 0 & 2 & 0 & 7 \end{bmatrix}$, K_4; (15) $\begin{bmatrix} 101 & 201 & 301 \\ 102 & 202 & 302 \\ 103 & 203 & 303 \end{bmatrix}$, $\begin{bmatrix} 1 & t & t^2 \\ t & 1 & t \\ t^2 & t & 1 \end{bmatrix}$; (16–17) rank-one A and skew-symmetric $K^T = -K$; (18) 3×3 Vandermonde $\det \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix} = (b-a)(c-a)(c-b)$.

20–34. 22–23: Find $\det(A - \lambda I) = 0$ for $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. **27:** Determinants of $\begin{bmatrix} 0 & a & 0 \\ 0 & 0 & b \\ c & 0 & 0 \end{bmatrix}$, 4×4 cyclic B, and $\begin{bmatrix} a & a & a \\ a & b & b \\ a & b & c \end{bmatrix}$. **30:** Show partial derivatives of $f(a,b,c,d) = \ln(ad - bc)$ give A^{-1}. **34:** If $\det A = 6$, find $\det B$ where rows of B are row 3 + row 2 + row 1, row 2 + row 1, row 1.

SOLUTION AT END OF PAGE (Problem Set 5.1 (Problems 1–34))

1. $\det(2A) = 2^4(\frac{1}{2}) = 8$, $\det(-A) = (-1)^4(\frac{1}{2}) = \frac{1}{2}$, $\det(A^2) = \frac{1}{4}$, $\det(A^{-1}) = 2$.
2. $\det(\frac{1}{2}A) = -\frac{1}{8}$, $\det(-A) = 1$, $\det(A^2) = 1$, $\det(A^{-1}) = -1$.
3. (a) **False**, (b) **True**, (c) **False** ($4^n|A|$), (d) **False**.
4–5. Number of row exchanges is $\lfloor n/2 \rfloor$ (or $\frac{n(n-1)}{2}$ adjacent swaps). For $n = 101$, $\lfloor 101/2 \rfloor =$

50 exchanges (even), so $\det J_{101} = \mathbf{+1}$.
6–12. 6: Factor $t = 0$ out of the zero row by Rule 3. **7:** $\det Q_{\text{rot}} = 1$, $\det Q_{\text{refl}} = -1$. **8:** $\det(Q^T Q) = (\det Q)^2 = 1 \implies \det Q = \pm 1$. **9:** $1, 2, 0$. **11:** $\det(-DC) = (-1)^n |D|\,|C|$, so for even n, $(-1)^n = +1$. **12:** A scalar factor $k = \frac{1}{ad-bc}$ multiplying a 2×2 matrix comes out squared: $k^2(ad - bc) = \frac{1}{ad-bc}$.
13–18. 13: 1 and 3. **14:** 12 and 5. **15:** 0 and $(1 - t^2)^2$. **16–17:** Both 0 (for odd n, $\det K = \det K^T = \det(-K) = (-1)^n \det K = -\det K \implies \det K = 0$). **18:** Subtract row 1 from rows 2, 3 and factor out $(b - a)(c - a)$ to get $(b - a)(c - a)(c - b)$.
22–34. 22: $(2 - \lambda)^2 - 1 = 0 \implies \lambda = 1, 3$. **23:** $\lambda^2 - 7\lambda + 10 = 0 \implies \lambda = 2, 5$. **27:** abc, $-abcd$ (needs 3 row exchanges), and $a(b - a)(c - b)$. **30:** $\frac{\partial f}{\partial a} = \frac{d}{ad-bc}, \frac{\partial f}{\partial c} = \frac{-b}{ad-bc}$, etc., matching A^{-1}. **34:** Subtracting rows leaves $\det B$ equal to $\det[\text{row } 3; \text{row } 2; \text{row } 1] = -\det A = \mathbf{-6}$.

4.2 Permutations and Cofactors [Book 5.2, pp. 255–268]

4.2.1 Derivation 10: Pivot Ratio Formula, Big Permutation Formula, and Cofactor Expansion

GOAL

Derive the three master formulas for $\det A$: (1) the Pivot Formula and Ratio $d_k = \det A_k / \det A_{k-1}$, (2) the Big Formula with $n!$ permutations, and (3) the Cofactor Formula of order $n-1$.

FROM YOUR +2 BASE

Use Rules 1–3 and mathematical induction on matrix size n.

Step 1: Pivots from Determinants of Upper-Left Corner Submatrices A_k. When $A = LU$ without row exchanges, the upper-left $k \times k$ submatrix factors into $A_k = L_k U_k$. By Rule 9 and Rule 7, $\det A_k = (\det L_k)(\det U_k) = d_1 d_2 \cdots d_k$. Dividing $\det A_k$ by $\det A_{k-1}$ isolates the kth pivot [Book: Eq. (3), p. 256]:

$$\boxed{\text{The } k\text{th pivot is } d_k = \frac{d_1 d_2 \cdots d_k}{d_1 d_2 \cdots d_{k-1}} = \frac{\det A_k}{\det A_{k-1}}}. \tag{4.4}$$

Step 2: Derive the Big Permutation Formula ($n!$ terms). Write each row of A as a sum of n coordinate row vectors: row $i = \sum_{\alpha=1}^{n} a_{i\alpha} \mathbf{e}_\alpha^T$. Apply Rule 3 (linearity in each of the n rows separately) to expand $\det A$ into n^n simple determinants. By Rule 4 (equal rows), any term that picks the same column twice is zero. Only the $n!$ terms that choose **one entry from each row and column** in a permuted column order $P = (\alpha, \beta, \ldots, \omega)$ survive [Book: Eq. (8), p. 258]:

$$\boxed{\det A = \sum_{\text{all } n! \text{ permutations } P} (\det P)\, a_{1\alpha} a_{2\beta} \cdots a_{n\omega} \qquad (\det P = \pm 1)}. \tag{4.5}$$

Step 3: Group terms by row i to get the Cofactor Expansion. In equation (4.5), collect all $(n-1)!$ terms that multiply a_{ij}. Since a_{ij} uses row i and column j, the remaining factors choose one entry from each remaining row and column, forming $\det M_{ij}$ (where M_{ij} is the $(n-1) \times (n-1)$ submatrix with row i and column j removed). Moving row i to row 1 takes $i-1$ swaps, and moving column j to column 1 takes $j-1$ swaps, giving the sign $(-1)^{(i-1)+(j-1)} = (-1)^{i+j}$ [Book: Eqs. (11)–(13), pp. 260–261]:

$$\textbf{Cofactor of } a_{ij}: \quad \boxed{C_{ij} = (-1)^{i+j} \det M_{ij}}, \tag{4.6}$$

$$\textbf{Expansion along row } i: \quad \boxed{\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}}, \tag{4.7}$$

$$\textbf{Expansion down column } j: \quad \boxed{\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}}. \tag{4.8}$$

4.2.2 Verbatim Book Examples and Worked Examples (Section 5.2)

BOOK EXAMPLE 5.2-Ex2&6 Determinant of the $-1, 2, -1$ Tridiagonal Matrix

Show by both the pivot formula and the cofactor recursion $D_n = 2D_{n-1} - D_{n-2}$ that the n by n tridiagonal $-1, 2, -1$ matrix has determinant $\det A_n = n + 1$.

BOOK EXAMPLE 5.2 A Hessenberg Matrix and Fibonacci Numbers

A *Hessenberg matrix* is a triangular matrix with one extra diagonal. Use cofactors of row 1 to show that the 4 by 4 determinant satisfies Fibonacci's rule $|H_4| = |H_3| + |H_2|$, where H_n has 2's on the main diagonal, 1's on the superdiagonal, and 1's on and below the main diagonal. Which Fibonacci number is $|H_n|$?

BOOK EXAMPLE 5.2 B Questions on the Big Formula

1. If A is the 10 by 10 all-ones matrix, how does the big formula give $\det A = 0$?
2. If you multiply all $n!$ permutations together into a single P, is P odd or even?
3. If you multiply each a_{ij} by the fraction i/j, why is $\det A$ unchanged?

SOLUTION AT END OF PAGE (Examples 5.2-Ex2&6, 5.2 A, and 5.2 B)

5.2-Ex2&6: (1) Pivots are $\frac{2}{1}, \frac{3}{2}, \frac{4}{3}, \ldots, \frac{n+1}{n}$, whose product telescopes to $\boxed{n+1}$. (2) Expanding along row 1 gives $D_n = 2D_{n-1} - (-1)(-1)D_{n-2} = 2D_{n-1} - D_{n-2}$. With $D_1 = 2, D_2 = 3$, induction gives $D_n = 2n - (n-1) = n+1$.

5.2 A: Expanding H_4 along row 1 gives $C_{11} = |H_3|$ and $C_{12} = -\det \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} = -(|H_3| - |H_2|)$. Thus $|H_4| = 2|H_3| - |H_3| + |H_2| = |H_3| + |H_2|$. Since $|H_1| = 2 = F_3$ and $|H_2| = 3 = F_4$, $\boxed{|H_n| = F_{n+2}}$.

5.2 B: (1) Every product in the big formula equals 1; half have $\det P = +1$ and half have $\det P = -1$, so they cancel to 0. (2) Odd for $n = 2, 3$; **even for all** $n \geq 4$ because the number of odd permutations is $n!/2$, which is an even integer for $n \geq 4$. (3) Each term $a_{1\alpha}a_{2\beta}\cdots a_{n\omega}$ is multiplied by $\frac{1\cdot 2\cdots n}{\alpha\cdot\beta\cdots\omega} = \frac{n!}{n!} = 1$, leaving $\det A$ unchanged.

4.2.3 Problem Set 5.2: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 5.2 (1–35) Big Formula, Cofactors, Recursions, and Block Determinants

1–2. Compute determinants from the 6 terms: (1) $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 4 & 4 \\ 5 & 6 & 7 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; (2) $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, C = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}$.

4. Find $\det A$ and $\det B$ for $A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{bmatrix}$.

5–6. 5: Smallest number of zeros in a 4×4 matrix to guarantee $\det A = 0$; maximum zeros while allowing $\det A \neq 0$. **6:** If $a_{11} = \cdots = a_{nn} = 0$, how many of the $n!$ terms are zero for $n = 3$ and $n = 4$?

9–10. 9: Show 4 is the largest determinant of a 3×3 (± 1)-matrix. **10:** Possible determinants of $I + P_{\text{even}}$ (4×4).

11–12. Find the cofactor matrix C and AC^T for $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 0 & 0 \end{bmatrix}$ and $K_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

13–16. Find recursions by cofactors for tridiagonal determinants: (13–14) C_n (0 on diagonal, 1s adjacent): find C_1, C_2, C_3, C_4 and C_{10}; (15) E_n (1, 1, 1 tridiagonal): show $E_n = E_{n-1} - E_{n-2}$ and find E_{100}; (16) F_n (1 on diagonal, -1 superdiagonal, 1 subdiagonal): show $F_n = F_{n-1} + F_{n-2}$.

19–22. 19: 4×4 Vandermonde V_4. **20:** All-ones minus identity $G_n = \text{ones}(n) - I_n$. **21–22:** 1, 3, 1 tridiagonal S_n: compute S_1, S_2, S_3, S_4.

23–35. 25: Prove $\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = |A|\,|D - CA^{-1}B| = |AD - CB|$ if $AC = CA$. **27:** Show $\frac{\partial(\det A)}{\partial a_{11}} = C_{11}$. **31:** Determinant of 4×4 cyclic permutation P. **32:** Prove $S_n = 3S_{n-1} - S_{n-2} = F_{2n+2}$. **33:** Why does subtracting 1 from the (n, n) entry of symmetric pascal(n) give $\det = 0$?

SOLUTION AT END OF PAGE (Problem Set 5.2 (Problems 1–35))

1–2. (1) $\det A = 12, \det B = 0, \det C = -1$ (rows of A, C independent). (2) $\det A = -2, \det B = 0, \det C = (-2)^2 = 4$.

4. $\det A = 1 - 1 = 0$ (rows 1 and 4 are equal); $\det B = (1)(3)(4)(2) - (2)(4)(4)(1) = 24 - 32 = -8$.

5–6. 5: 4 zeros (one full row/col) guarantees $\det A = 0$; 12 zeros (leaving 4 nonzeros on diagonal) allows $\det A \neq 0$. **6:** For $n = 3$: 4 terms are zero (leaving 2 derangements). For $n = 4$: 15 terms are zero (leaving 9 derangements).

9–10. 9: In 6 terms of ± 1, the product of all 6 is -1, so they cannot all be $+1$; at most five are $+1$ and one is -1, giving $5 - 1 = 4$. **10:** $\det(I + P_{\text{even}}) \in \{0, 4, 16\}$.

11–12. For B: $\det B = 7(12 - 15) = -21$. For K_3: $C = C^T = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ and $K_3 C^T = 4I$.

13–16. (13–14) $C_n = -C_{n-2}$ with $C_1 = 0, C_2 = -1, C_3 = 0, C_4 = 1 \implies C_{10} = -1$. (15) $E_1 = 1, E_2 = 0, E_3 = -1, E_4 = -1, E_5 = 0, E_6 = 1$ (period 6) $\implies E_{100} = E_4 = -1$. (16) $F_n = F_{n-1} + F_{n-2} =$ Fibonacci F_{n+1}.
19–22. 19: Zero at $x = a, b, c$. **20:** $\det G_n = (-1)^{n-1}(n-1)$ ($G_2 = -1, G_3 = 2, G_4 = -3$). **21:** $S_1 = 3, S_2 = 8, S_3 = 21, S_4 = 55$ (F_{2n+2}).
25–33. 25: Take determinants of $\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}$. **27:** In $\det A = a_{11}C_{11} + \cdots + a_{1n}C_{1n}$, none of the cofactors C_{1j} depend on a_{11}, so $\frac{\partial(\det A)}{\partial a_{11}} = C_{11}$. **31:** 3 exchanges $\implies \det P = -1$, $|P^2| = 1$. **33:** In $P = LU$, the (n, n) pivot is 1; subtracting 1 from P_{nn} reduces the last pivot in U from 1 to 0, so det $= 0$.

4.3 Cramer's Rule, Inverses, and Volumes [Book 5.3, pp. 269–282]

4.3.1 Derivation 11: Cramer's Rule, Cofactor Inverse $A^{-1} = C^T/\det A$, Volumes, and Cross Product

GOAL

Derive Cramer's Rule $x_j = \det B_j/\det A$, the Cofactor Inverse Formula $A^{-1} = C^T/\det A$, the Determinant Formula for Box Volume and Jacobians, and the 3D Cross Product and Scalar Triple Product.

FROM YOUR +2 BASE

Use Rule 9 ($\det(AB) = (\det A)(\det B)$), **T6** (2D triangle area), and **T8** (3D cross product).

Step 1: Derive Cramer's Rule for $A\mathbf{x} = \mathbf{b}$. Replace column j of the identity matrix I by the unknown vector $\mathbf{x}$. By Example 4 of Section 5.2, the determinant of this modified identity matrix is x_j. Multiply A times this matrix a column at a time: column j becomes $A\mathbf{x} = \mathbf{b}$, while all other columns remain columns of A, forming the matrix B_j [Book: Eqs. (1)–(4), p. 269]:

$$A\begin{bmatrix}\mathbf{e}_1 & \dots & \mathbf{x} & \dots & \mathbf{e}_n\end{bmatrix} = \begin{bmatrix}\mathbf{a}_1 & \dots & \mathbf{b} & \dots & \mathbf{a}_n\end{bmatrix} = B_j. \tag{4.9}$$

Take determinants using Rule 9 to get $(\det A)(x_j) = \det B_j$:

$$\boxed{\textbf{Cramer's Rule:} \quad x_j = \frac{\det B_j}{\det A} \qquad (B_j \text{ has column } j \text{ of } A \text{ replaced by } \mathbf{b})}. \tag{4.10}$$

Step 2: Derive the Cofactor Formula for A^{-1}. The jth column of A^{-1} solves $A\mathbf{x} = \mathbf{e}_j$. By Cramer's Rule, its ith entry $(A^{-1})_{ij}$ has $\mathbf{b} = \mathbf{e}_j$ placed into column i of A. Expanding that determinant along column i gives the cofactor $C_{ji} = (-1)^{j+i}\det M_{ji}$. Equivalently, multiplying row i of A by the cofactors of row i gives $\det A$, while multiplying row i of A by the cofactors of a *different* row $k \neq i$ gives 0 (the determinant of a matrix with two equal rows) [Book: Eqs. (6)–(8), pp. 270–271]:

$$\boxed{AC^T = (\det A)I \implies A^{-1} = \frac{1}{\det A}C^T \qquad \text{or} \qquad (A^{-1})_{ij} = \frac{C_{ji}}{\det A}}. \tag{4.11}$$

Step 3: Area, n-Dimensional Box Volume, and Calculus Jacobians. Consider an n-dimensional box whose edges from the origin are the rows of A. (1) When $A = I$, the box is the unit cube with volume $1 = \det I$ (Rule 1). (2) Exchanging two rows leaves the same box and the same positive volume $|\det A|$ (Rule 2). (3) Scaling row 1 by t scales the box length by t, and adding edge $1'$ to edge 1 (with height and other edges fixed) adds the volumes (Rule 3). Because box volume obeys Rules 1–3, we have:

$$\boxed{\text{Volume of Box with Edges in Rows (or Columns) of } A = |\det A|}. \tag{4.12}$$

In 2D, a triangle with corners $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is half of a parallelogram [Book: Eq. (9), p. 272]:

$$\boxed{\text{Area of Triangle} = \frac{1}{2}\left|\det\begin{bmatrix}x_1 & y_1 & 1\\ x_2 & y_2 & 1\\ x_3 & y_3 & 1\end{bmatrix}\right| = \frac{1}{2}|x_1y_2 - x_2y_1| \text{ when } (x_3, y_3) = (0,0)}. \tag{4.13}$$

In polar coordinates $x = r\cos\theta, y = r\sin\theta$, the infinitesimal box area is $\mathrm{d}A = J\,\mathrm{d}r\,\mathrm{d}\theta$, where the **Jacobian** is:

$$\boxed{J = \begin{vmatrix} \partial x/\partial r & \partial x/\partial \theta \\ \partial y/\partial r & \partial y/\partial \theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r}. \tag{4.14}$$

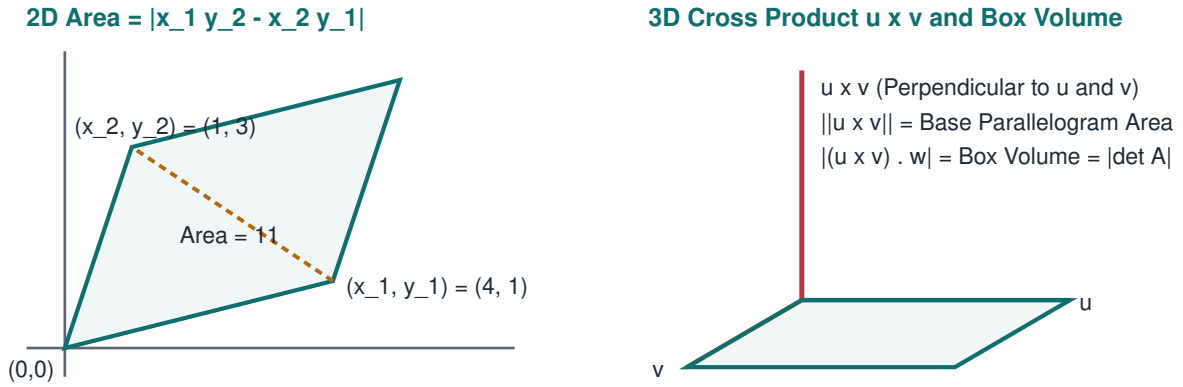


Fig. 4.1 Left: Parallelogram area $|x_1y_2 - x_2y_1| = 11$ and triangle area $11/2$. Right: 3D cross product $\mathbf{u} \times \mathbf{v}$ and scalar triple product $|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = \text{Volume} = |\det A|$.

Step 4: Cross Product $\mathbf{u} \times \mathbf{v}$ and Scalar Triple Product $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$. For $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ in $\mathbb{R}^3$, expand the symbolic 3×3 determinant along row 1 [Book: Eqs. (10)–(13), pp. 275–276]:

$$\boxed{\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2)\hat{\mathbf{i}} + (u_3v_1 - u_1v_3)\hat{\mathbf{j}} + (u_1v_2 - u_2v_1)\hat{\mathbf{k}}}, \tag{4.15}$$

$$\boxed{(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \pm(\text{Volume of } \mathbf{u}, \mathbf{v}, \mathbf{w} \text{ box})}. \tag{4.16}$$

4.3.2 Verbatim Worked Examples (Section 5.3)

BOOK EXAMPLE 5.3 A Finding the Nullspace of a Singular Matrix by Cofactors

If A is singular, $AC^T = (\det A)I$ becomes $AC^T =$ **zero matrix**. *Then each column of C^T is in the nullspace of A.* Solve $A\mathbf{x} = \mathbf{0}$ by $\mathbf{x} =$ cofactors along a row, for these singular matrices of rank 2:

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 3 & 9 \\ 2 & 2 & 8 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

BOOK EXAMPLE 5.3 B Cramer's Rule, Cofactor Inverse, and Box Volumes

Use Cramer's Rule with ratios $\det B_j / \det A$ to solve $A\mathbf{x} = \mathbf{b}$ for $\begin{bmatrix} 2 & 6 & 2 \\ 1 & 4 & 2 \\ 5 & 9 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

Also find $A^{-1} = C^T / \det A$, and find the volumes of the boxes whose edges are columns of A and rows of A^{-1}.

SOLUTION AT END OF PAGE (Worked Examples 5.3 A and 5.3 B)

5.3 A: For the first matrix, the cofactors along row 1 are $C_{11} = 24 - 18 = 6$, $C_{12} = -(16 - 18) = 2$, $C_{13} = 4 - 6 = -2$, giving $\mathbf{x} = (6, 2, -2)^T \in N(A)$. For the second matrix, the cofactors of row 1 are all 0, so use row 2 to get $C_{21} = -(1 - 2) = 1$, $C_{22} = 1 - 2 = -1$, $C_{23} = 0 \implies \mathbf{x} = (1, -1, 0)^T \in N(A)$.

5.3 B: The determinant of A is $\det A = 5(12 - 8) - 9(4 - 2) = 2$. Cramer's Rule replaces each column by $\mathbf{b} = (0, 0, 1)^T$: $\det B_1 = 1(12 - 8) = 4 \implies x = 4/2 = 2$; $\det B_2 = -1(4 - 2) = -2 \implies y = -2/2 = -1$; $\det B_3 = 1(8 - 6) = 2 \implies z = 2/2 = 1$. Thus $\mathbf{x} = (2, -1, 1)^T$. The cofactor matrix is $C = \begin{bmatrix} -18 & 10 & -11 \\ 18 & -10 & 12 \\ 4 & -2 & 2 \end{bmatrix}$, so $A^{-1} = \frac{C^T}{\det A} = \begin{bmatrix} -9 & 9 & 2 \\ 5 & -5 & -1 \\ -11/2 & 6 & 1 \end{bmatrix}$. The third column of A^{-1} matches $\mathbf{x}$. Volume of the column box is $|\det A| = \boxed{2}$. Volume of the A^{-1} row box is $|\det(A^{-1})| = \frac{1}{|\det A|} = \boxed{1/2}$.

4.3.3 Problem Set 5.3: Verbatim Exercises with End-of-Page Solutions

BOOK EXAMPLE PS 5.3 (1–18) Cramer's Rule, Inverses, and Area/Volume

1. Solve by Cramer's Rule $\det B_j / \det A$: $2x_1 + 3x_2 = 1, 4x_1 + 5x_2 = 7$.

2. Solve $x - y = 2, x + y = 4$ by Cramer's Rule and check by inspection.

3. Use Cramer's Rule to solve for y without finding x, z, t:

$$\begin{aligned} x + 4y - z + t &= 2 \\ 2x + 7y + z - t &= 1 \\ -x - 2y - z + 2t &= 3 \\ 3x + 10y + 2z - t &= 4. \end{aligned}$$

4. Compute x_1, x_2, x_3 by Cramer's Rule for $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

5. If all entries of A and $\mathbf{b}$ are integers, and $\det A = 1$, prove that the solution $\mathbf{x}$ has integer components.

6. Find the cofactor matrix C and the inverse $A^{-1} = C^T / \det A$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$.

7. Invert $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}$ using cofactors.

8. If A is symmetric ($A^T = A$), show from $C_{ij} = (-1)^{i+j} \det M_{ij}$ that the cofactor matrix C is also symmetric ($C^T = C$).

9. If $\det A = 1$, what is $\det C$? What is the inverse of the cofactor matrix C?

10. For a 3×3 matrix with $\det A = 2$, find $\det C$, $\det(C^T)$, and $\det(A^{-1})$.

11. Show that the (i, j) entry of A^{-1} is zero if and only if the cofactor $C_{ji} = 0$.

12. If A is a tridiagonal matrix, explain why the $(1,3)$ and $(3,1)$ entries of A^{-1} are usually not zero, even though $a_{13} = a_{31} = 0$.

13. Find the area of the parallelogram with vertices $(0,0), (2,1), (1,3), (3,4)$.

14. Find the area of the triangle with corners $(1,2), (3,6), (4,5)$.

15. Find the volume of the box (parallelepiped) whose edges from the origin are the rows of $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$.

16. A tetrahedron has vertices at $(0,0,0)$ and the three rows of A. Explain why its volume is $\frac{1}{6}|\det A|$.

17. Find the volume of the tetrahedron with vertices $(0,0,0), (1,0,0), (1,1,0), (1,1,1)$.

18. If the three columns of A have lengths L_1, L_2, L_3, prove Hadamard's inequality: $|\det A| \leq L_1 L_2 L_3$, with equality if and only if the columns are mutually orthogonal.

SOLUTION AT END OF PAGE (Problem Set 5.3 (Problems 1–18))

1. $\det A = 10 - 12 = -2$; $\det B_1 = 5 - 21 = -16 \implies x_1 = 8$; $\det B_2 = 14 - 4 = 10 \implies x_2 = -5$.

2. $\det A = 2$; $\det B_1 = 6 \implies x = 3$; $\det B_2 = 2 \implies y = 1$.

3. $y = \det B_2 / \det A$, where B_2 replaces column 2 by $(2,1,3,4)^T$. Elimination yields $\det A = -2$ and $\det B_2 = -2$, so $y = 1$.
4. $\det A = -2$; $\det B_1 = -1 \implies x_1 = 1/2$; $\det B_2 = -1 \implies x_2 = 1/2$; $\det B_3 = 1 \implies x_3 = -1/2$.
5. By Cramer's Rule, $x_j = \det B_j / \det A = \det B_j / 1 = \det B_j$. Since B_j has integer entries, its determinant is an integer.
6. $\det A = 24$; $C = \begin{bmatrix} 24 & 0 & 0 \\ -12 & 6 & 0 \\ -2 & -5 & 4 \end{bmatrix}$; $A^{-1} = \frac{C^T}{24} = \begin{bmatrix} 1 & -1/2 & -1/12 \\ 0 & 1/4 & -5/24 \\ 0 & 0 & 1/6 \end{bmatrix}$.
7. $\det A = -1$; $C = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & -1 \\ 1 & -2 & 1 \end{bmatrix} \implies A^{-1} = \frac{C^T}{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix}$.
8. Transposing A transposes each submatrix M_{ij}, giving $\det M_{ji} = \det M_{ij}$, so $C_{ji} = C_{ij} \implies C^T = C$.
9. From $AC^T = (\det A)I = I$, $C^T = A^{-1}$. Thus $\det C = \det A^{-1} = 1$, and $C^{-1} = A^T$.
10. $AC^T = 2I \implies (\det A)(\det C^T) = 2^3 = 8 \implies \det C^T = \det C = 4$. $\det(A^{-1}) = 1/\det A = 1/2$.
11. By equation (4.11), $(A^{-1})_{ij} = C_{ji} / \det A$, which vanishes if and only if $C_{ji} = 0$.
12. The minor $M_{31} = \det \begin{bmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{bmatrix} = \det \begin{bmatrix} a_{12} & 0 \\ a_{22} & a_{23} \end{bmatrix} = a_{12}a_{23} \neq 0$ when the subdiagonals are nonzero.
13. Area $= |\det \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}| = |6 - 1| = \mathbf{5}$.
14. Area $= \frac{1}{2}|\det \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ 4 & 5 & 1 \end{bmatrix}| = \frac{1}{2}|-7| = \mathbf{7/2}$.
15. Volume $= |\det A| = 1 \cdot 4 \cdot 6 = \mathbf{24}$.
16. A pyramid has volume $\frac{1}{3}$(base area)(height). The triangular base has area $\frac{1}{2}$(parallelogram area), giving $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}|\det A|$.
17. Volume $= \frac{1}{6}|\det \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}| = \frac{1}{6}(1) = \mathbf{1/6}$.
18. Factor $A = QR$ where Q has orthonormal columns ($\det Q = 1$) and R is upper triangular with $r_{ii} \leq \|\text{col } i\| = L_i$. Thus $|\det A| = |\det R| = \prod r_{ii} \leq L_1 L_2 L_3$, with equality iff R is diagonal (A has orthogonal columns).

BOOK EXAMPLE PS 5.3 (19–38) Cross Products, Triple Products, Jacobians, and Challenge Problems

19. Compute $\mathbf{u}\times\mathbf{v}$ for $\mathbf{u}=(1,2,3)$ and $\mathbf{v}=(4,5,6)$. Check that $\mathbf{u}\cdot(\mathbf{u}\times\mathbf{v})=0$ and $\mathbf{v}\cdot(\mathbf{u}\times\mathbf{v})=0$.

20. Show from the determinant definition that $\mathbf{u}\times\mathbf{v}=-(\mathbf{v}\times\mathbf{u})$ and $\mathbf{u}\times\mathbf{u}=\mathbf{0}$.

21. Verify Lagrange's identity $\|\mathbf{u}\times\mathbf{v}\|^2+(\mathbf{u}\cdot\mathbf{v})^2=\|\mathbf{u}\|^2\|\mathbf{v}\|^2$ using $\|\mathbf{u}\times\mathbf{v}\|=\|\mathbf{u}\|\,\|\mathbf{v}\|\sin\theta$.

22. Find a unit vector $\mathbf{w}$ perpendicular to both $\mathbf{u}=(1,1,1)$ and $\mathbf{v}=(1,-1,0)$.

23. Compute the box volume $|(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}|$ for $\mathbf{u}=(1,1,0),\mathbf{v}=(0,1,1),\mathbf{w}=(1,0,1)$.

24. If $\mathbf{u},\mathbf{v},\mathbf{w}$ lie in the same plane, explain why $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}=0$.

25. Prove the vector triple product formula $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})=(\mathbf{u}\cdot\mathbf{w})\mathbf{v}-(\mathbf{u}\cdot\mathbf{v})\mathbf{w}$.

26. Find the 3×3 Jacobian matrix J and its determinant for cylindrical coordinates $x=r\cos\theta, y=r\sin\theta, z=z$.

27. Find the Jacobian determinant for spherical coordinates $x=\rho\sin\phi\cos\theta, y=\rho\sin\phi\sin\theta, z=\rho\cos\phi$. Show $J=\rho^2\sin\phi$.

28. If each entry of $A(t)$ is differentiable, show that $\frac{\mathrm{d}}{\mathrm{d}t}(\det A)=\sum_{i,j}\frac{\mathrm{d}a_{ij}}{\mathrm{d}t}C_{ij}=\operatorname{tr}\left(C^T\frac{\mathrm{d}A}{\mathrm{d}t}\right)$.

29. Use Problem 28 to find $\frac{\mathrm{d}}{\mathrm{d}t}(\det A)$ when $A(t)=\begin{bmatrix}e^t & t\\ t^2 & \cos t\end{bmatrix}$.

30. For the 2×2 differential equation $\mathbf{y}'=A\mathbf{y}$, show that the Wronskian $W(t)=\det[\mathbf{y}_1(t)\ \ \mathbf{y}_2(t)]$ satisfies Liouville's formula $\frac{\mathrm{d}W}{\mathrm{d}t}=(\operatorname{tr}A)W(t)$.

31. What is the volume of a 4-dimensional box whose edges are the columns of $A=2I_4$?

32. If A is $n\times n$ with rows $\mathbf{r}_1,\dots,\mathbf{r}_n$, show that $|\det A|\le\prod_{i=1}^{n}\|\mathbf{r}_i\|$.

33. Find the 3×3 matrix M such that $M\mathbf{v}=\mathbf{u}\times\mathbf{v}$ for fixed $\mathbf{u}=(u_1,u_2,u_3)$. Show that $M^T=-M$ (skew-symmetric).

34. Show that the cross product matrix M in Problem 33 satisfies $M^3=-\|\mathbf{u}\|^2M$.

35. If $A\mathbf{x}=\mathbf{0}$ for a 3×3 matrix with rank 2, explain why any non-zero row of the cofactor matrix C gives a solution $\mathbf{x}$.

36. Use cofactors to find a nonzero vector perpendicular to the rows of $A=\begin{bmatrix}1&2&3\\2&3&4\end{bmatrix}$.

37. Show that the area of the polygon with vertices $(x_1,y_1),\dots,(x_n,y_n)$ in counterclockwise order is $\frac{1}{2}\sum_{i=1}^{n}(x_iy_{i+1}-x_{i+1}y_i)$ with $(x_{n+1},y_{n+1})=(x_1,y_1)$.

38. (Challenge) Prove that the derivative of $f(t)=\det(e^{At})$ at $t=0$ is the trace of A, and deduce Jacobi's formula $\det(e^A)=e^{\operatorname{tr}(A)}$.

SOLUTION AT END OF PAGE (Problem Set 5.3 (Problems 19–38))

19. $\mathbf{u}\times\mathbf{v}=(-3,6,-3)^T$. Dot products: $1(-3)+2(6)+3(-3)=0$ and $4(-3)+5(6)+6(-3)=0$.

20. Exchanging rows 2 and 3 of the symbolic determinant reverses its sign (Rule 2). Identical rows give zero (Rule 4).

21. $\|\mathbf{u}\times\mathbf{v}\|^2+(\mathbf{u}\cdot\mathbf{v})^2=\|\mathbf{u}\|^2\|\mathbf{v}\|^2(\sin^2\theta+\cos^2\theta)=\|\mathbf{u}\|^2\|\mathbf{v}\|^2$.

22. $\mathbf{u}\times\mathbf{v}=(1,1,-2)^T$; unit vector $\mathbf{w}=\frac{1}{\sqrt{6}}(1,1,-2)^T$.

23. $\det\begin{bmatrix}1&1&0\\0&1&1\\1&0&1\end{bmatrix}=1(1)-1(-1)=\mathbf{2}$.

24. $\mathbf{u}\times\mathbf{v}$ is normal to the plane of $\mathbf{u}$ and $\mathbf{v}$. If $\mathbf{w}$ lies in that plane, $\mathbf{w}$ is perpendicular to $\mathbf{u}\times\mathbf{v}$, so the dot product is zero.

25. Both sides are linear in $\mathbf{u},\mathbf{v},\mathbf{w}$. Setting $\mathbf{u}=\hat{\mathbf{i}},\mathbf{v}=\hat{\mathbf{i}},\mathbf{w}=\hat{\mathbf{j}}$ gives $\mathbf{0}=\mathbf{0}$; setting

$\mathbf{u} = \hat{\mathbf{j}}, \mathbf{v} = \hat{\mathbf{i}}, \mathbf{w} = \hat{\mathbf{j}}$ gives $-\hat{\mathbf{i}} = -\hat{\mathbf{i}}$. By linearity, the identity holds for all vectors.

26. $J = \det\begin{bmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = r(\cos^2\theta + \sin^2\theta) = \mathbf{r}$.

27. Expansion along row 3 gives $J = \cos\phi(\rho^2\cos\phi\sin\phi) - (-\rho\sin\phi)(\rho\sin^2\phi) = \rho^2\sin\phi(\cos^2\phi + \sin^2\phi) = \boldsymbol{\rho^2\sin\phi}$.

28. Expand $\det A$ along row i: $\det A = \sum_j a_{ij}C_{ij}$. Differentiating gives $\sum_{i,j} a'_{ij}C_{ij} = \operatorname{tr}(C^T A')$. If A is invertible, $C^T = (\det A)A^{-1}$, so $(\det A)' = (\det A)\operatorname{tr}(A^{-1}A')$.

29. $\frac{\mathrm{d}}{\mathrm{d}t}(\det A) = \det\begin{bmatrix} e^t & 1 \\ t^2 & \cos t \end{bmatrix} + \det\begin{bmatrix} e^t & t \\ 2t & -\sin t \end{bmatrix} = (e^t\cos t - t^2) + (-e^t\sin t - 2t^2) = \mathbf{e^t}(\cos\mathbf{t} - \sin\mathbf{t}) - \mathbf{3t^2}$.

30. $W'(t) = \det[\mathbf{y}_1' \;\; \mathbf{y}_2] + \det[\mathbf{y}_1 \;\; \mathbf{y}_2'] = \operatorname{tr}(A)\det[\mathbf{y}_1 \;\; \mathbf{y}_2] = (\operatorname{tr} A)W(t)$.

31. $\det(2I_4) = 2^4 = \mathbf{16}$. **32.** Same as Hadamard's inequality applied to A^T: $|\det A| = |\det A^T| \le \prod \|\mathbf{r}_i\|$.

33. $M = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}$, which satisfies $M^T = -M$.

34. Characteristic equation of M is $\det(M - \lambda I) = -\lambda(\lambda^2 + \|\mathbf{u}\|^2) = 0$. By the Cayley-Hamilton theorem, $M^3 + \|\mathbf{u}\|^2 M = 0 \implies M^3 = -\|\mathbf{u}\|^2 M$.

35. Since $\det A = 0$, $AC^T = (\det A)I = 0$, so every column of C^T (every row of C) satisfies $A\mathbf{x} = \mathbf{0}$.

36. Cross product of the two rows: $(1,2,3) \times (2,3,4) = (-1,2,-1)^T \in N(A)$.

37. Decompose the polygon into triangles with the origin as a common vertex. The signed area of the ith triangle is $\frac{1}{2}(x_i y_{i+1} - x_{i+1}y_i)$. Summing over all triangles gives the total area.

38. At $t = 0$, $e^{At} = I$ and $\frac{\mathrm{d}}{\mathrm{d}t}(e^{At}) = A$. By Problem 28, $f'(0) = (\det I)\operatorname{tr}(I^{-1}A) = \operatorname{tr}(A)$. Since $e^{A(t+s)} = e^{At}e^{As}$, $f(t+s) = f(t)f(s)$, which implies $f(t) = e^{ct}$ with $c = f'(0) = \operatorname{tr}(A)$. At $t = 1$, $\det(e^A) = e^{\operatorname{tr}(A)}$.

Chapter 5: Unique Synthesis Problems & Formula Sheet

Chapter 5 Core Equations (Derived from +2 Base)

- **Defining Axioms (Rules 1–3):** $\det I = 1$, row exchange reverses sign ($\det P = \pm 1$), linear in each row separately ($\det(tA) = t^n \det A$).
- **Pivot Formula:** $\det A = \pm(d_1 d_2 \cdots d_n) = \pm \det U$; kth pivot is $d_k = \det A_k / \det A_{k-1}$.
- **Big Permutation Formula:** $\det A = \sum_P (\det P)\, a_{1\alpha} a_{2\beta} \cdots a_{n\omega}$ ($n!$ terms, $n!/2$ positive and $n!/2$ negative).
- **Cofactor Formula:** $C_{ij} = (-1)^{i+j} \det M_{ij}$; expansion along row i: $\det A = \sum_{j=1}^{n} a_{ij} C_{ij}$.
- **Matrix Inverse via Cofactors:** $A^{-1} = \dfrac{1}{\det A} C^T$, where $AC^T = (\det A) I$.
- **Cramer's Rule:** $x_j = \dfrac{\det B_j}{\det A}$ (B_j replaces column j with $\mathbf{b}$).
- **Volume & Area:** nD Box Volume $= |\det A|$; 2D Triangle Area $= \frac{1}{2} \left| \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right|$; Tetrahedron Volume $= \frac{1}{6} |\det A|$.
- **Cross & Triple Products:** $\mathbf{u} \times \mathbf{v} = \det \begin{bmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix}$; Box Volume $= |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = |\det[\mathbf{u}\ \mathbf{v}\ \mathbf{w}]|$.
- **Matrix Derivative & Jacobi Formula:** $\dfrac{\mathrm{d}}{\mathrm{d}t}(\det A) = (\det A) \operatorname{tr}\left(A^{-1} \dfrac{\mathrm{d}A}{\mathrm{d}t}\right)$, $\det(e^A) = e^{\operatorname{tr}(A)}$.

BOOK EXAMPLE Ch5-U1 Unique Synthesis Problem: Continuous Determinant and Wronskian

(a) For the $n \times n$ matrix $A(t) = I + tB$, use the big formula to show that $\frac{\mathrm{d}}{\mathrm{d}t} \det(I + tB)\Big|_{t=0} = \operatorname{tr}(B)$.
(b) Derive Jacobi's identity $\det(e^B) = e^{\operatorname{tr}(B)}$ by setting $f(t) = \det(e^{tB})$ and solving the differential equation $f'(t) = \operatorname{tr}(B) f(t)$ with $f(0) = 1$.
(c) For second-order ODE $y'' - 4y' + 3y = 0$ with basis solutions $y_1(x) = e^x$ and $y_2(x) = e^{3x}$, compute the Wronskian determinant $W(x) = \det \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}$ and verify Abel-Liouville's formula $W(x) = W(0) \exp\left(-\int_0^x (-4)\, \mathrm{d}t\right)$.

SOLUTION AT END OF PAGE (Chapter 5 Unique Synthesis Problem Ch5-U1)

(a) In $\det(I + tB) = \sum_P (\det P) \prod_{i=1}^{n} (I + tB)_{i,P(i)}$, the identity permutation $P = (1, 2, \ldots, n)$ gives the diagonal product:

$$(1 + tb_{11})(1 + tb_{22}) \cdots (1 + tb_{nn}) = 1 + t(b_{11} + b_{22} + \cdots + b_{nn}) + O(t^2) = 1 + t \operatorname{tr}(B) + O(t^2).$$

Every non-identity permutation picks at least two off-diagonal entries, contributing $O(t^2)$.

Taking $\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}$ gives $\boxed{\operatorname{tr}(B)}$.

(b) Since $e^{(t+h)B} = e^{tB}e^{hB} = e^{tB}(I + hB + O(h^2))$, Rule 9 gives:

$$f(t+h) = \det(e^{(t+h)B}) = \det(e^{tB})\det(I + hB + O(h^2)) = f(t)\big(1 + h\operatorname{tr}(B) + O(h^2)\big).$$

Subtracting $f(t)$ and dividing by $h \to 0$ gives $f'(t) = \operatorname{tr}(B)f(t)$. The unique solution with $f(0) = \det I = 1$ is $\boxed{f(t) = e^{t\operatorname{tr}(B)}}$. At $t = 1$, $\boxed{\det(e^B) = e^{\operatorname{tr}(B)}}$.

(c) The Wronskian is:

$$W(x) = \det\begin{bmatrix} e^x & e^{3x} \\ e^x & 3e^{3x} \end{bmatrix} = 3e^{4x} - e^{4x} = \boxed{2e^{4x}}.$$

From Abel's formula with $P(x) = -4$ and $W(0) = 2$:

$$W(x) = W(0)\exp\left(-\int_0^x (-4)\,\mathrm{d}t\right) = 2\exp(4x) = \boxed{2e^{4x}},$$

which exactly verifies Abel-Liouville's formula.